

Review of field-reversed configurations

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Loren C. Steinhauer



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Review of field-reversed configurations

Loren C. Steinhauer^{a)}

Department of Aeronautics and Astronautics, University of Washington, Seattle, Washington 98195, USA

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This review addresses field-reversed configurations (FRCs), which are compact-toroidal magnetic systems with little or no toroidal field and very high β (ratio of plasma pressure to magnetic pressure). Although enthusiasm for the FRC has primarily been driven by its potential for an attractive fusion reactor, this review focuses on the physics rather than on technological or engineering aspects. Major advances in both theory and experiment have taken place since the previous comprehensive FRC review in 1988. Even so many questions remain. In particular, even though FRC experiments have exhibited remarkable stability, how well this extrapolates to larger systems remains unresolved. The review considers FRCs under familiar topical categories: equilibrium, global stability, self-organization, transport, formation, and sustainment. © 2011 American Institute of Physics. [doi:10.1063/1.3613680]

I. INTRODUCTION

A. Background

The field-reversed configuration (FRC) is now over fifty years of age. It emerged as early as 1956 with the Astron concept. It was proposed that an energetic electron layer of sufficient density might reverse the magnetic field, creating a closed field region propitious for good confinement.¹ The first supposed appearance of field reversal in an actual experiment was portrayed by a drawing (Fig. 1) of what early researchers supposed had occurred in θ -pinches.^{2,3} Even so, they expressed little hope that this peculiar arrangement of plasma and magnetic field might have any enduring use. Yet, these early glimmers marked the beginning of a saga, long and varied, with multiple ups and downs, which has continued to this day.

FRCs have been reviewed before; the earliest, in 1982, focused on a particular manifestation.⁴ The most comprehensive was Tuszewski's massive 1988 review,⁵ which remains the best survey of "early" FRCs. The most recent review, in 1996, focused largely on programmatic issues.⁶ In the decade and a half since, the understanding of FRC physics has advanced considerably, and notable innovations have appeared. The present review mainly addresses progress since 1988. It focuses on the physics with only scant mention of the technology of FRC experiments. Further, the acclaimed merits of an FRC-based fusion reactor^{7,8} are only mentioned briefly.

B. Definition and uniqueness

The FRC is a toroidal magnetic confinement system with little or no toroidal field, composed of a set of nested toroidal magnetic surfaces.⁹ It belongs to the family of compact toroids, where "compact" denotes the absence of a central column. As such, the plasma extends all the way to the geometric axis. (A looser definition allows a central column so long as it does not house the inner leg of a toroidal

field coil.) Figure 2 illustrates the basic anatomy of an FRC. The *separatrix* forms the nominal boundary, separating closed from open magnetic surfaces. The *O-point* or magnetic axis is the innermost surface, shrunk to a ring. The primary dimensions are the separatrix radius R_s and length L_s . As in the figure, FRC studies usually employ cylindrical coordinates (r, θ, z) . Also shown is a conducting wall at radius R_c , which performs the same role as the equilibrium field coil set in tokamaks. Following is a brief listing of features that are unique as viewed from the perspectives of both engineering and physics.

a. Engineering. The primary appeal of the FRC as a fusion system is technological. (1) *Geometric simplicity*: the configuration is linear; the absence of a center column makes for easier maintenance of a fusion core. (2) *Magnetic simplicity*: an open magnet system with no toroidal field coils; this may also eliminate the specter of damaging disruptions. (3) *High- β* : i.e., high pressure and thus high power density in a fusion system; this allows the possibility of advanced, aneutronic fusion fuels. (4) *Natural divertor*: the open magnet system allows the scrape-off-layer exhaust to be collected outside the confinement region with no area restriction on the divertor plate. (5) *Movability*: the FRC is a *moveable* object; it can be formed in an independent "source" chamber and translated into a separate confinement chamber.

b. Physics. Although classed as a toroidal magnetic system, the FRC possesses physical properties quite fantastical compared to other such systems. (1) *Extreme β* because of the relatively small magnetic field in the core; this gives rise to physics not captured by the standard magnetohydrodynamic (MHD) model. (2) *Large-orbit effects* follow from high- β ; indeed, near the O-point and X-points, the Larmor radius is infinite. (3) *Dominant drifts*: the flows are primarily *across* the magnetic field with ions carrying most of the current. (4) *Large flow shear* appears naturally near the separatrix. (5) *Isolation*: the separatrix is quite remote from walls

^{a)}Electronic mail: lstein@uw.edu.

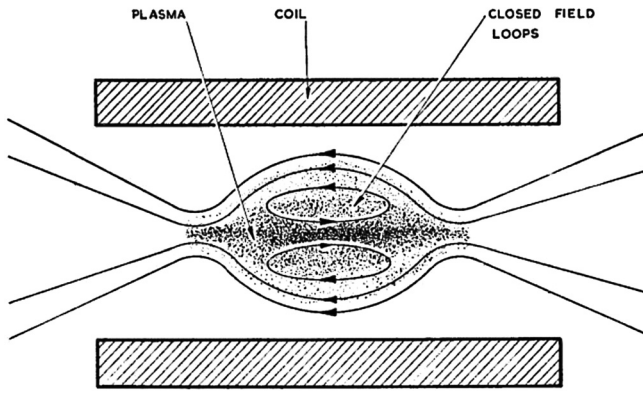


FIG. 1. Early conceptual FRC. Reprinted with permission from J. K. Wright and N. J. Phillips, *J. Nucl. Energy, Part C* **1**, 240 (1960). Copyright 1960, IOP Publishing.

with separation distance of 2–3 times the minor radius in many experiments.

c. Essential parameters. Three parameters are commonly used to describe the FRC geometry. (1) *Radial size parameter.* The most practical of these is

$$S_* \equiv R_s / (c / \omega_{pi}) \quad (1)$$

where c / ω_{pi} is the ion skin depth and $\omega_{pi} = (e^2 n / \epsilon_0 m_i)^{1/2}$ is the ion plasma frequency. (Refer to the Appendix for essential FRC terminology in common use.) In typical experiments, $S_* \sim 3 - 30$. Two other radial size parameters have also been used, both based on the ion Larmor radius rather than the ion skin depth.¹⁰ The reciprocal of the size parameters measures the importance of two-fluid and finite-Larmor-radius effects. (2) *Separatrix elongation:*

$$E \equiv L_s / 2R_s \quad (2)$$

Observe that this is the elongation (length-to-diameter ratio) of the separatrix, *not* that of individual magnetic surfaces (κ) commonly used in tokamaks. The elongation affects global stability. *Oblate* FRCs have $E < 1$; *prolate* or “elongated” FRCs have $E > 1$. FRCs formed (Re Sec. VI) by spheromak merging typically have $E \sim 0.5 - 1$, while those formed by θ -pinch methods range from $E \sim 3 - 30$. (3) *Relative radius*, a measure of the separation between the FRC and the side boundary:

$$X_s \equiv R_s / R_c. \quad (3)$$

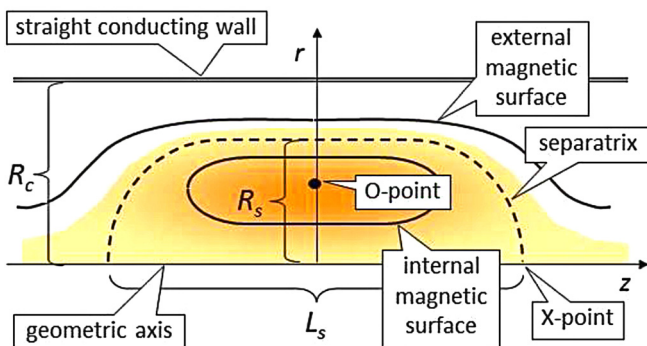


FIG. 2. (Color online) Anatomy of an FRC.

This measure is only used in elongated FRCs where it has a strong effect on the amount of closed poloidal flux ϕ_p . Typical FRCs formed by θ -pinch methods have $X_s = 0.3 - 0.6$, while those formed by rotating magnetic fields (Re Sec. VI) have $X_s = 0.9$.

d. Role in plasma science. The FRC is a promising fusion concept claiming attractive engineering and economic benefits. Moreover, FRC research benefits the development of other magnetic fusion concepts as well as the understanding of magnetospheric and astrophysical plasma physics. Indeed, it offers a unique test bed for exploring phenomena missing from the standard MHD model: multi-fluids, high- β , large-ion-orbit effects, relaxation to “natural” states, and reconnection.

C. Brief history

a. Paleo-FRCs. The first likely appearance of FRCs was in θ -pinches in 1960,^{2,3} although the label “FRC” did not emerge for more than two decades. An early researcher² predicted that the closed magnetic flux structure would rapidly disappear because of instability and high dissipation rate; this proved prescient. In several experiments during the 1960s, FRCs exhibited two readily observable instabilities (tearing, rotational, Re Sec. III) that destroyed the configuration within 10 μ s, which is little better than the free-flowing end-loss time in an ordinary θ -pinch. This brought the demise of the concept, even though one experiment indicated that the tearing instability, at least, could be eliminated.¹¹ Meanwhile through the 1970s, Soviet researchers quietly continued experiments of θ -pinch FRCs. Reporting in 1978, they made the spectacular claim of complete stability with lifetimes many times the end-loss time.¹² Key to this achievement was the abandonment of the 1960s push for large neutron yield in favor of a more physics-based approach. This included more care with the formation “recipe,” and the use of much more modest magnetic fields and compressions ($X_s > 0.4$, as opposed to 0.1 – 0.2 in early FRCs). In the same era, others began to consider the possibility of reversing the magnetic field in a mirror machine.^{13,14} This concept became known as a field-reversed mirror (FRM), which is essentially an FRC sustained by neutral beam injection.

b. Resurgence. Word of the Soviet success sparked modest FRC programs in the United States and Japan. This was part of the mushrooming interest in “compact toroids” (CT), ultra-low aspect ratio toroidal plasmas without toroidal field coils. This includes spheromaks, which have a substantial internal toroidal field, and FRCs, which have little or none. A large CT symposium at Princeton in 1979 launched a long series of symposia, many of which were joint meetings with Japanese scientists. Thus began the heyday of FRC research that would last throughout the 1980s. The label “FRC” for field-reversed configuration appeared early in this era, probably first appeared in print in 1982.^{4,15} This era produced FRCs larger and longer-lived than any until recently. Further, the θ -pinch formation method matured, benefiting

from several improvements. In parallel, a variety of theoretical tools emerged to deal with the highly nonstandard plasma properties of FRCs.

c. Second collapse and re-emergence. In 1990, economic and programmatic pressures came to bear, leading to the collapse of FRC programs in the United States and the former Soviet Union. Despite this second virtual extinction elsewhere, two Japanese FRC facilities continued and a third was added. In the mid to late 1990's FRC research resumed in the United States under the heading of "innovative confinement concepts" (ICC). The first of a series of ICC workshops was at Princeton in 1998. This latest era continues although daunting economic and programmatic pressures continue. While not generally produced more impressive parameters (e.g., S_* , temperature), this era is notable for the appearance of new formation methods (Re Sec. VI), for various theory advances, and more recently the appearance of a substantial private sector investment in FRC research.^{16,17} In summary, despite two collapses, FRC research has re-emerged phoenix-like, each time in new and innovative forms.

d. Current experiments. Quite a number of FRC facilities have operated over the last five decades. Those before 1988 are listed in Ref. 5. Table I lists of current experiments, i.e., those active in the past five years. (An asterisk denotes facilities that have ceased operation.) The category "type"

specifies the formation method. Table II displays typical parameters reported for these facilities.

D. Fundamental issues

a. Emerging FRC concept. Experiments in recent years have shaken some traditional pictures of the FRC. Following are several and where they will be discussed. (1) *Toroidal field*: some FRCs have a significant toroidal field, with a substantial safety factor, yet with high- β (Sec. II). (2) *MHD*: that the standard MHD model poorly describes FRC stability is well known; it is also a poor model of FRC equilibrium (Sec. II). (3) *Ion current*: most FRCs are ion-current dominated as a natural property of the equilibrium (Sec. II). (4) *Relaxation*: FRCs are high- β "natural states" maintained by continuous relaxation processes (Sec. IV). (5) *Ruggedness*: FRCs are remarkably robust objects that can survive formation methods involving capture and collision at highly super-Alfvenic speeds (Sec. VI).

b. Summary of physics issues. While FRCs have been exploited for investigating physics relevant to space plasmas (e.g., reconnection), the primary driver of FRC research has been its fusion energy potential. The issues that follow relate to fusion. (1) *Equilibrium*. Nearly, all modeling of FRC equilibria has assumed a single-fluid MHD, which performs poorly in the edge region (Sec. II) where two-fluid and kinetic

TABLE I. Current FRC experiments.

Facility	Type	Institution	References
<i>Tokyo Spheromak-3</i>			
TS-3	merging-spheromak	Univ. Tokyo	18, 19
<i>Nihon University Compact Torus Experiment-III</i>			
NUCTE-III	θ -pinch	Nihon Univ.	20, 21
<i>FRC Injection eXperiment*</i>			
FIX	translation-trapping	Osaka Univ.	22
<i>Magnetic Reconnection eXperiment</i>			
MRX	spheromak merging	Princeton Plasma Phys. Lab.	23
<i>Translation Confinement Sustainment*</i>			
TCS	rotating magnetic fields	Univ. Washington	24, 25
<i>Swarthmore Spheromak eXperiment</i>			
SSX	merging-spheromak	Swarthmore Coll.	26
<i>Field-Reversed eXperiment-Liner</i>			
FRX-L	θ -pinch	Los Alamos Natl. Lab.	27
<i>Tokyo Spheromak-4</i>			
TS-4	merging-spheromak	Univ. Tokyo	28, 29
<i>Translation Confinement Sustainment-Upgrade*</i>			
TCS-U	rotating magnetic fields	Univ. Washington	30, 31
<i>Princeton Field Reverse eXperiment</i>			
PFRC	rotating magnetic fields	Princeton Plasma Phys. Lab.	32
<i>Plasma Liner Compression</i>			
PLC	translation-trapping	Mathematical Sci. Northwest	33, 34
<i>Inductive Plasma Accelerator</i>			
IPA	collision-merging	Mathematical Sci. Northwest	35
<i>Colorado</i>	spheromak merging	Univ. Colorado	36
<i>Irvine Field-Reversed Configuration</i>			
IFRC	coaxial source	Univ. California Irvine	37
C-2	collision-merging	Tri Alpha Energy	16, 17

TABLE II. Typical parameters in FRC experiments.

Facility	$n_e(10^{19}\text{m}^{-3})$	$T_i(\text{eV})$	$R_s(\text{cm})$	E
TS-3	5 – 10	20 – 100	20	1
NUCTE-III	5 – 400	200 – 300	5 – 7	5 – 10
FIX	3 – 5	200 – 300	15 – 20	7 – 15
MRX	5 – 20	30	50	0.3 – 0.7
TCS	0.1 – 2.5	25 – 50	35 – 37	–
SSX	100	20 – 40	20	1.5
FRX-L	1500 – 2500	200 – 700	1.5 – 2.5	7 – 10
TS-4	–	10 – 40	60 – 70	0.5 – 0.7
TCS-U	0.4 – 1.5	50 – 200	35 – 37	1.5 – 3
PFRC	0.05 – 0.3	200 – 300	3	–
IPA	23 – 26	350	10	–
PLC	–	–	10	–
IFRC	150	10	30	–
C-2	5 – 10	200 – 500	30 – 40	3 – 5

effects are important. (2) *Stability*. The evident stability of elongated FRCs remains unexplained, although contributing factors have been identified: e.g., finite Larmor radius and sheared flow. Stability at large- S^* remains unclear and awaits a better understanding of these non-standard MHD effects. (3) *Confinement*. The understanding of transport in FRCs remains at an empirical, non-physics based level. This includes the cross-field transport of particles and energy, flow in the scrape-off layer, and the rate of poloidal-flux dissipation. The mechanisms driving these various elements of global confinement must be characterized before confident extrapolations to fusion can be made. (4) *Sustainment*. While several current drive and heating methods have been investigated, their relevance to large, high-temperature FRCs is questionable. A viable sustainment method remains to be developed. (5) *Formation*. FRCs now benefit from a range of formation methods, some of which might extrapolate to a fusion system. Even so, further development is needed, especially in the context of larger, higher magnetic field systems.

c. Conclusion. The FRC concept is old, even predating the tokamak. This raises the obvious question—why then is the FRC much less developed. The simplistic answer is that progress has been limited and does not yet merit a large development program. This is partly true. A more complete explanation recognizes the plain fact that the FRC is fantastical compared to conventional fusion concepts, so much so that only an optimistic vision can conceive it as a practical fusion concept. Further, its physics is so unique that one only imagines with difficulty how FRC research contributes to the development of other concepts. Even so, FRC development is presently in its *third* life, having passed twice through immolation and rising from the ashes. The promise of a much more attractive fusion concept motivates these resurgences, and so FRC research continues.

d. Outline. The section outline of the remainder of this review is organized along the following topical lines: (II) Equilibrium, (III) Stability, (IV) Self-organization, (V) Transport, (VI) Formation technology, and (VII) Sustainment.

II. EQUILIBRIUM

A. The challenge of FRC equilibria

Section I broadly announced the unusual nature of the FRC among magnetic confinement systems. Moving to the topic of equilibrium and looking more closely, even more peculiarities appear. (1) All FRCs have *extremely high* β . In a tokamak, $\langle\beta\rangle$ (average plasma pressure $\div$ average magnetic pressure) of 0.1 is considered high. By contrast, in a typical FRC, the $\langle\beta\rangle$ defined in the same way is an enormous, 8–12.³⁸ Indeed the common measure of β employed for FRCs is actually an *engineering* $\beta \equiv p/(B_e^2/2\mu_0)$, i.e., based on B_e , the external magnetic field supplied by the magnet. Defined in this way, typical values lie in the range $\langle\beta\rangle \sim 0.8 - 0.95$. (2) The flows are *predominantly drifts*, a byproduct of extremely high- β . (3) *Large flow shear* appears naturally near the separatrix. (4) *Strong kinetic effects* arise throughout the interior, but especially near the O-point and X-points, where the magnetic field vanishes, and near the separatrix where the SOL thickness is comparable to the ion Larmor radius. The kinetic nature of the X-point region is enhanced because it is a true point, not a ring as in a tokamak. (5) Turbulence produces *electrically shorted magnetic surfaces* throughout the interior as well as the open field region. In addition to the foregoing, some FRCs have additional unique equilibrium properties. (6) θ -pinch-formed FRCs have *extreme elongation* (Re Sec. VI) e.g., as portrayed in Fig. 2. Thus, from the outside they present a “linear” rather than a conventional “toroidal” geometry, an attribute with celebrated engineering advantages. (7) θ -pinch-formed FRCs are *extremely isolated*, i.e., quite distant from wall boundaries even on the sides. The typical ratio of external-to-internal poloidal flux is an enormous $\phi_e/\phi_p = 20 - 40$.³⁹ This makes such FRCs more like space plasmas than any other magnetic confinement concept.

The foregoing unique features present considerable challenges to understanding FRC equilibria. Common approximations fail, e.g., low β , and large aspect ratio = major-to-minor radius. Further, standard and even extended-fluid models cannot capture highly kinetic properties. Finally, it is experimentally difficult to “get inside” an FRC diagnostically, especially to detect the internal field structure. Many interesting FRCs are simply too hot and dense to allow internal magnetic probes.

The remainder of this section discusses basic equilibrium models, surveys detailed features of FRC equilibria and finally summarizes experimental results.

B. Equilibrium models

By definition, a plasma equilibrium is a steady state condition where momentum sources or sinks (e.g., friction forces) can be neglected. This is always an approximation since dissipative forces are always present to some degree. However, if the dynamical timescale, e.g., the Alfvén time $\tau_A \sim R_s/V_A$ is short compared with the time scale associated, e.g., the resistive time $\tau_{res} = \mu_0 R_s^2/\eta$, then equilibrium is a useful construct. In fact, the Lundquist number, which is the ratio of these two $Lu = \tau_{res}/\tau_A$ is large in all FRC

experiments. The theory of plasma equilibria is a *hierarchy* of descriptions with increasing sophistication, which can be classified under the headings static-fluid, flowing-fluid, and kinetic.

a. Static fluid equilibrium. i.e., stationary ions, is the “base” model. Here, the “massless” electrons are the only current carriers. The Grad-Shafranov (GS) equation for the magnetic flux function $\psi(r, z)$ is

$$\Delta^* \psi = -I(\psi)I'(\psi) - \mu_0 r^2 p'(\psi), \quad (4)$$

where $\Delta^* = r^2 \nabla \cdot [(1/r^2) \nabla]$ is the GS operator; the two surface functions are $I(\psi) = rB_\theta$ and $p(\psi)$, and the prime denotes the derivative with respect to ψ . In a traditional FRC, the toroidal field is negligible, $I(\psi) \approx 0$, leaving only one “arbitrary” function. Numerical methods for solving the GS equation have long been in wide use, and numerical difficulties associated with elongated FRCs were addressed and managed satisfactorily in the 1980s. However, analytic or quasi-analytic static equilibrium for everyday use in interpreting experiments has continued value. Static FRCs investigated in recent years include the following. Numerical GS solvers have commonly been used in support of stability studies^{40–45} and in parallel with experimental observations.^{46–49} A simplified GS solution based on the uniformly elongated flux-surface approximation yielded useful information about the internal two-dimensional (2D) structure^{50,51} despite exhibiting unlikely features in the SOL and jet. The gyroviscosity offers a fluid-based estimate of the finite-Larmor-radius effect.⁵² FRC stability studies have included this correction (Re Sec. III),^{53–55} but no equilibrium model as yet.

b. Flowing fluid equilibria. The ions can flow steadily and still satisfy the basic definition of equilibrium. With such flows the ion inertia may become important. An early study addressed FRC equilibria with rigid ion rotational flow.⁵⁶ However, the full multi-fluid formulation did not appear until 1999.⁵⁷ It highlighted unique features of multi-fluid, flowing equilibria. (a) Each fluid species has its own set of characteristic surfaces. These are $\psi = \text{const}$ for massless electrons, but each ion fluid species “ j ” has its own distinct family $P_{j0} = \text{const}$, where

$$P_{j0} = q_j \psi + m_j r u_{j0} \quad (5)$$

is the canonical fluid momentum. (b) Each species has *three* “arbitrary” surface functions: the “total” enthalpy, entropy, and poloidal flow stream function. (c) The equilibrium system for each species is a second-order partial-differential equation plus a scalar Bernoulli equation. (d) While the single-fluid MHD description has no characteristic length, a multi-fluid has one for each ion species, the ion skin depth c/ω_{pi} . In the two-fluid context (ions + electrons), this introduces a size parameter $S_* \equiv R_s/(c/\omega_{pi})$. Recently, the multi-fluid model was extended to allow collisionless ions;⁵⁸ in this case, the entropy surface function is replaced by one for the ion temperature.

Several unusual features appear in the two-fluid model. *Complex separatrix:* the ions and electrons have distinct separatrices, i.e., $\psi = 0$ differs from $P_{j0} = 0$.⁵⁹ This causes closed and open surface effects to overlap because the ion fluid confinement boundary ($P_{j0} = 0$) extends into the open magnetic field region ($\psi > 0$). Ions in the transitional region are magnetically confined while electrons are electrostatically confined by ambipolar potentials. The single-fluid flowing model, which has only $\psi = \text{const}$ surfaces, completely misses this feature. Thus far, two-fluid equilibrium solutions have received limited study. An early study investigated “stationary-energy” equilibria (Re Sec. IV).⁶⁰ Others found equilibria using a uniform density approximation, but only in the context of a spherical torus.⁶¹ A quasi-one-dimensional solver for elongated FRCs was developed and applied to interpret an experiment.⁶²

c. Kinetic equilibria. The Vlasov equation governs the distribution function for a population of collisionless particles. Any solution to the steady-state ($\partial/\partial t = 0$) Vlasov equation can be expressed as a function of the constants of motion. In an axisymmetric configuration there are two constants, the Hamiltonian $H_j = (1/2)m_j v_j^2 + q_j \phi$ and the canonical momentum P_{j0} [Eq. (5) with u_{j0} replaced by v_{j0}] where $\mathbf{v}_j$ is the individual ion velocity. The most widely used analytic paradigm for FRC equilibrium is, in fact, kinetic. The “rigid rotor” (RR) equilibrium, originally derived to describe θ -pinch equilibria,⁶³ also applies to FRCs. In the RR, the distribution function has the form $F \propto \exp(-H + \Omega P_\theta)$. For modest ion rotation, this leads to an analytic solution to be described shortly under the heading “analytic equilibria.”

Other than the popular RR equilibrium, fully kinetic descriptions of FRC-like equilibria have rarely been employed and always in the context of systems with large-ion orbits, e.g., the ion-ring system (Astron) (Refs. 64–67) or a hybrid system with an ion ring imbedded in a pressure-bearing background plasma.⁶⁸ These models usually assume the RR distribution, although the Astron work modified this to account for prompt end-loss of ions with unconfined orbits.^{64,65} More recently, preliminary results appeared for a different kind of hybrid equilibrium model: kinetic ions (one or more species) with massless fluid-electrons.⁶⁹ Turning to simulations, some have combined a fluid treatment for the “thermal” ions and a kinetic treatment of energetic “beam” ions, as might be produced by neutral beam injection. The first such FRC-relevant model was developed to investigate the stabilizing effect of the beam.⁷⁰ Other simulations have investigated the influence of the beam ions on equilibrium *per se*.^{71,72} Other kinetic studies have focused on ion orbits: identifying the confinement boundary,⁷³ the effect of the self-consistent electric field,^{74–76} orbit classification for high-energy fusion products,⁷⁷ and in the presence of a rotating magnetic field.⁷⁸

d. Analytic models. Analytic descriptions of FRC equilibria have proved useful for interpreting experiments, as well as presenting a simple base for stability and transport studies. FRCs have very low aspect ratio (major radius $\div$ minor radius) so that toroidal effects on equilibrium

are significant. Thus, high-aspect-ratio approximations (e.g., the “bicycle tire”) do not apply. As a result, only one fully 2D analytical equilibrium exists, the so-called Hill’s vortex:

$$\psi = -(3/4)B_e r^2 [1 - (r^2 + z^2)/R_s^2] \quad r^2 + z^2 \leq R_s^2, \quad (6a)$$

$$\psi = -(1/2)B_e r^2 [1 - R_s^3/(r^2 + z^2)^{3/2}] \quad r^2 + z^2 > R_s^2, \quad (6b)$$

where B_e is the “external field” (B_z) far outside the separatrix. The difficulty with this simple result is its inflexibility: it is an isolated solution with only one separatrix elongation $E=1$, only one pressure function $p(\psi) = -15B_e\psi/2\mu_0 R_s^2$ for $\psi \leq 0$ and no scrape-off layer (SOL), i.e., $p(\psi)=0$ for $\psi > 0$. A slightly more useful 2D solution is the “Solov’ev” equilibrium which has arbitrary-elongation version: replace z^2/R_s^2 in Eq. (6a) by $z^2/E^2 R_s^2$. Unfortunately, it only applies inside the separatrix.

A much greater breadth of equilibria are available for elongated FRCs, which apply if E exceeds 2 or 3. As such they are quasi-one-dimensional with weak dependence on z over much of their length. The most well-known example is the widely used RR profile:

$$B_z = B_e \tanh(Ku) \quad n = n_m \text{sech}^2(Ku), \quad (7)$$

where n_m is the maximum density, $u \equiv 2r^2/R_s^2 - 1$ is the minor radius variable, and K is the “shape” parameter. Equation (7) applies for small ion rotation rate (negligible centrifugal effect); a slightly more complicated form includes the centrifugal effect. The advantage of the RR over the Hill’s vortex is the appearance of the “shape” parameter K . Consequently, it contains a *family* of pressure profiles $p(\psi) = (B_e^2/2\mu_0) \text{sech}^2 K \times \exp(-8K\psi/B_e R_s^2)$. However, the apparent freedom to choose K is a false one. As will be seen in Sec. II C (under “axial force balance”), K is set by other factors. An even greater defect of the RR equilibrium is its poor representation of the external field region; the SOL is much too broad.

Various more flexible 1D equilibria have appeared using multiple parameters.^{79–82} A middle ground between too few

and too many parameters is “two,” suggesting the “two-point equilibrium” (2PE) model.^{83,84} This recognizes that FRCs have *two* fundamental length scales, the separatrix radius, R_s , and the SOL thickness, L_n . The ratio of these is comparable to $S_* \sim 5$ –30. Figure 3 shows the weakness of the RR, which has a very broad SOL ($r/R_s > 1$) compared to the realistic 2PE. Even more marked is the difference between the current density profiles (Fig. 3(b)). The “truncated RR” is occasionally used, in which n drops abruptly to zero at $r=R_s$, eliminating the SOL altogether, but corresponding to a current density spike at the edge.

Improved 2D analytic equilibria are also of interest, particularly to allow a more flexible separatrix shape than the “elliptical” shape $r^2/R_s^2 + z^2/Z_s^2 = 1$, where $Z_s = \pm L_s/2$ are the X-point locations. The separatrix may be more racetrack-like with nearly constant separatrix radius vs z along most of the length. One such approximately matches analytic interior and exterior solutions with a flexible separatrix shape.⁸⁵ Another analytic solution focuses only on the interior.⁸⁶ Finally, a “quasi-equilibrium” was constructed to allow a modest toroidal field component.⁸⁷

C. Experimental measurements of equilibria

A variety of diagnostics have been applied in FRCs, although the available methods vary depending on the geometry and constraints imposed by the formation method. The basic diagnostic suite routinely used in low-elongation merging-spheromak formed FRCs includes (a) internal magnetic probe arrays and (b) impurity line emission and broadening. The basic diagnostic suite in θ -pinch-based formation methods includes (a) single-chord interferometry measuring $\int n dl$ across the diameter and (b) an excluded-flux array measuring the excluded flux is $\Delta\phi = B_e \pi R_c^2 - \phi(R_c)$ as a function of z . The excluded flux radius is $R_{\Delta\phi} = (R_c^2 - \Delta\phi/B_e \pi)^{1/2}$. It has long been assumed that $R_{\Delta\phi} \approx R_s$. Recent interpretive modeling found this to be accurate within 2%–3%.^{49,84} The average density can then be inferred from these measurements as $\bar{n} \approx \int n dl / 2R_{\Delta\phi}$. The maximum density n_m is a few percent higher than $\bar{n}$.

Internal magnetic probing cannot be used in high-temperature, high-density θ -pinch-formed FRCs but has been used in lower-density elongated FRCs: FRX-C/T,⁸⁸ FIX,²² TCS (translation-trapping mode),^{62,87,89} and TCS (rotating-magnetic-field mode).^{90,91} Some elongated experiments have employed multi-chord interferometry $\int n dl$ vs y (impact parameter) and used an Abel inversion to infer the radial density profile $n(r)$. These include the moderate density FRX-C/T (Ref. 80) and PIACE (Ref. 92) and the high-density FRX-L (Ref. 27, 93) experiments. One θ -pinch formed experiment, NUCTE-III (Ref. 47) inserted a single, on-axis magnetic probe to measure the field structure near the X-point. Other experiments in NUCTE-III measured multi-chord bremsstrahlung emission, approximately $\int n^2 dl$ vs y , allowing extraction of information about the density profile.^{46,47}

D. Details of FRC equilibria

a. Radial force balance. Several features are noteworthy, most of which apply only to elongated FRCs. One is

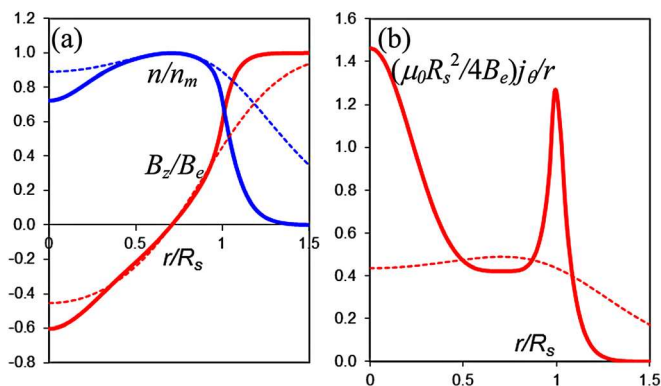


FIG. 3. (Color online) Profiles of density, magnetic field, and current density (j_θ/r) of a 2PE equilibrium (solid lines) and a rigid rotor (dashed lines) for a typical $R_s/R_c = 0.4$, $S_* = 15$, $L_n/(c/\omega_{pi}) = 2$ (Fig. 1 of Ref. 84, adapted).

radial force balance, often called “pressure balance,” which applies for $E > 2$ (negligible curvature force at the midplane) and negligible toroidal field:

$$p + B_z^2/2\mu_0 = B_e^2/2\mu_0, \quad (8)$$

This allows the internal magnetic field structure to be inferred if the pressure structure can be found in some way. Radial force balance has proved to be an extremely useful property. Since $B_z = 0$ at the O-point, the “pressure balance temperature” $T_i = B_e^2/2\mu_0 n_m k$ is commonly inferred.

b. Axial force balance. Another property of elongated FRCs is *axial force balance*, which for a fixed-radius flux conserver at $r = R_c$ leads to the extremely simple expression, the so-called “average- β relation”⁹⁴

$$\langle \beta \rangle = 1 - R_s^2/2R_c^2, \quad (9)$$

The average $\langle \beta \rangle$ is taken inside the separatrix at the midplane and is the “engineering” β based on the external field. For the “truncated RR” profile, Eq. (9) leads to $K \approx (3/2)^{1/2} R_s/R_c$, i.e., K is not a free parameter. A shortcoming of the average- β relation is that it only applies for a fixed radius flux conserver. Some experiments, however, apply strong magnetic mirrors at the ends. A set of GS computations for mirror ratio up to four explored this effect.⁹⁵ A warning is in order: occasionally Eq. (9) has been misapplied to cases where the FRC “leans on” such magnetic mirrors. Equation (9) also ignores the SOL and the rocket effect of outflow in the jet. A model correcting for these effects⁸⁴ showed Eq. (9) to be a good approximation for $X_s \geq 1/2$ but increasingly inaccurate for lower X_s .

Combining radial and axial force balance has fostered approximations for the internal magnetic flux. These have proved useful since internal magnetic field measurements are unavailable in hot, high density, FRCs. Define the flux profile factor as $f_\phi \equiv \phi_p/B_e \pi R_s^2$: the RR profile gives $f_\phi = 0.31X_s$; an expression derived in Ref. 5 is $f_\phi \approx 0.32X_s^{5/4}$; and another based on the 2PE paradigm⁸⁴ has $f_\phi \approx 0.27X_s^{2/3}$. For a typical ratio $X_s = 0.4$, these give 0.12, 0.10, and 0.15, respectively.

c. Radial structure. The current profile at the midplane is *hollow* if j_θ/r has a minimum at the O-point and *peaked* if it has a maximum there. The 2PE example in Fig. 3(b) has a hollow current profile. An MHD simulation showed a mild peak of j_θ/r near the O-point and large spike at the separatrix,⁴² cp Fig. 3(b). Quasi-analytic modeling considered both peaked and hollow profiles.⁵¹ The spontaneous appearance of a hollow current profile appeared in a kinetic equilibrium simulation.⁹⁶ Interpretive transport modeling of experiment has been used to infer the current profile;^{97,98} this showed that the “current profile index” has a close correlation with X_s .⁹⁷ It is unclear whether this results from a self-organization process⁹⁷ or is merely an equilibrium property of elongated FRCs.⁵⁰

d. Axial structure. Other studies have addressed the separatrix shape. It can be characterized as $r^2 R_s^2 + (z/Z_s)^\sigma = 1$, where $\sigma = 2$ for an elliptic and $\sigma > 2$ for a “racetrack” separa-

trix. GS solutions suggested that racetrack shapes are more likely for larger elongation E .⁸⁵ Other GS solutions indicated that peaked $p'(\psi)$ led to an elliptic separatrix, while flat or hollow $p'(\psi)$ led to a racetrack.⁵¹ Interpretation of excluded-flux array results from FRX-C suggested a roughly elliptic shape.⁹⁹ Near-infrared radiation intensity measurements (axial array) on NUCTE-III indicated “elliptical” for lower E and “racetrack” for larger E .⁴⁶

e. Multiple O-points. The classical FRC has a single O-point, as in Fig. 1, but multiple O-points are also a possibility, e.g., a doublet structure, appeared in 2D-MHD simulations of formation.¹⁰⁰ Multiple “internal magnetic islands” were inferred from interpretive modeling of excluded flux array observations.⁴⁷ Single- and double-null 2D analytic equilibria were found in some GS solutions.⁸⁶

f. Toroidal field. While the classical picture of an FRC has a purely poloidal field, a modest toroidal field may be present without compromising the very high β . A kinetic simulation of FRC formation and stability showed a persistent “up-down” toroidal field (TF) structure, with positive TF on one end and negative on the other.¹⁰¹ The maximum magnitude of the TF was $\sim 15\%$ of the external field so that β was still high. Simulation of the SSX experiment also exhibited an up-down TF that did not completely annihilate.¹⁰² Internal field measurements in TCS operated in a translation-trapping mode (Re Sec. VI) (Ref. 87, 103) indicated a TF with maximum magnitude about 30% of the external field. While this is a modest TF, quasi-equilibrium modeling showed that the safety factor was roughly 1/2 at the O-point and exceeded two near the separatrix, a consequence of the large elongation.

In experiment, internal field probe arrays have been applied in FRCs formed using translational methods (Re Sec. VI). These have observed significant toroidal fields up to 30% of the external field: FRX-C/T,⁸⁸ FIX,²² and TCS (translation-trapping mode).^{62,87,89}

g. Applied external field structure. “Vertical” field coils are the common method for shaping the plasma separatrix in tokamaks. In FRCs, little attention has been given to field shaping. For example, elongated FRCs generally employ a fixed-radius flux conserver which sets the separatrix shape. However, some studies have considered external field modifications. A set of GS solutions investigated the effect of strong mirrors at the ends of the flux conserver.⁹⁵ In FIX, active mirrors were applied to shorten L_s and to increase R_s .^{104,105} NUCTE-III introduced gradual external field gradients, i.e., “field-index” control.¹⁰⁶ A rotating-magnetic-field experiment used shaping coils to produce a “doublet” like separatrix profile.¹⁰⁷

h. Spontaneous equilibrium flow and flow shear FRCs have strong rotational flow: $u_{i\theta}$ is the sum of the electric (“E $\times$ B”) drift $V_E = \hat{\theta} \cdot (\mathbf{E} \times \mathbf{B})/B^2$ and (in the fluid description) the ion-diamagnetic drift $V_{i*} = \hat{\theta} \cdot (\mathbf{B} \times \nabla p_i)/q_i n B^2$. The ion rotation is often referenced to V_{i*} , and in many analyses, it is simply assumed to be a rigid $u_{i\theta}/r = \Omega_i = \text{const}$. If the rotation is prescribed, the electrical potential ($\mathbf{E} = -\nabla \phi$)

“floats” in response to it. On the other hand, it is commonly understood that the region outside the separatrix is subject to end-shortening^{108,109} causing it to rotate at roughly the ion diamagnetic speed. A recent Hall-MHD simulation accounted for the shorting effect at a wall boundary.¹¹⁰

Understanding the core rotation is important because it drives rotational instabilities (Re Sec. III). Several studies have sought the cause of core rotation, i.e., “spin-up;” all but one of these are based on dissipative effects. The non-dissipative mechanism, “electrical shorting” is discussed here, and the dissipative mechanisms will be discussed shortly under the heading “spin-up”

The time scale for end-shortening on open field lines $\tau_{\text{short}}(\text{SOL}) \sim L_s/2V_{A0}$, where $L_s/2$ is a typical connection length to an external shorting boundary. This is a relatively short time in typical FRC experiments ($0.3 - 5 \mu\text{s}$). Shorting between magnetic surfaces is likely to occur in the core region as well as a result of normal turbulence. The original idea of broken magnetic surfaces is an old one¹⁵ raised in the context of the current drive by neutral beam injection. It re-emerged lately as an explanation for rotation in spheromaks¹¹¹ as well as FRCs.⁸³ Electromagnetic turbulence breaks magnetic surfaces, causing field lines to wander. Electrons have high mobility along field lines and rapidly make electrical contact between magnetic surfaces. The effect is to “short out” the core region, cancelling and even slightly reversing^{83,109} the inward electric field. If so, then electrical shorting occurs not just in the external region but throughout the FRC. The shorting time in the core is longer than the SOL shorting time $L_s/2V_{A0}$, but only moderately so since the connection length from the core to the periphery is at most a few times $L_s/2$, a consequence of the low internal equilibrium magnetic field.^{83,111} In summary, the shorting of the internal electric field generates a spontaneous rotation not directly dependent on a frictional effect. The rotational flow is comparable to the ion diamagnetic drift V_{i*} . Further, while surface breaking allows electrical shorting and anomalously rapid electron thermal conduction, charge neutrality prevents a catastrophic “end loss” of electrons, which are confined by the ambipolarity potential.

While the RR has uniform rotation frequency $u_{i0}/r \sim V_{i*}/r = \text{const}$, this is simply not so in realistic equilibria. Indeed, the flow is highly sheared near the separatrix. Spontaneous ion rotation with strong flow shear appears in MHD (Ref. 42) as well as kinetic^{45,96,102,112} simulations and in interpretive modeling of FRC equilibria.⁸⁴ Figure 4 shows the flow profile from the MHD simulation. Spontaneous flow shear has implications for stability and transport (Re Secs. III and V).

Spectroscopic techniques have occasionally been used to infer rotational flows. In the low-elongation, TS-3 (Ref. 113) rotational flow ($\sim 10 \text{ km/s}$) appeared simultaneous with the merging-formation and persisted into the equilibrium. The SSX experiment,¹¹⁴ on the other hand, observes little equilibrium flow, which may be a consequence of the plasma boundary lying directly on the wall and high viscosity. A few early spectroscopic measurements were made in elongated FRCs,⁵ and more recently on NUCTE-III (Ref. 115) and TCS.^{25,62} Some of these flows exhibit moderate to large flow shear. In TS-3, the flow rate was maximum near the

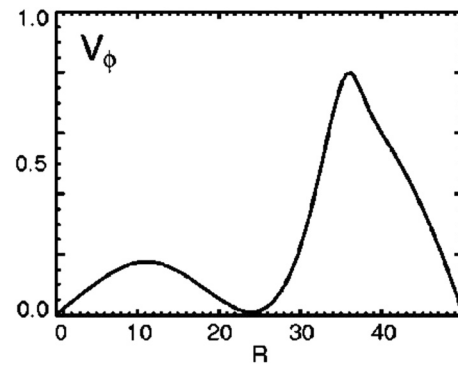


FIG. 4. Rotational flow in an MHD simulation (Fig. 3(b) of Ref. 42); V_θ is normalized by V_{A0} and the separatrix radius is $R_s \approx 35$ (cm). Reprinted with permission from E. V. Belova, S. C. Jardin, H. Ji, M. Yamada, and R. Kulsrud, Phys. Plasmas **7**, 4996 (2000). Copyright 2000, American Institute of Physics.

O-point and fell toward the edge.¹¹³ Unfolded spectroscopic measurements on TCS indicate large flow shear⁶² with the largest shear appearing inside the O-point. Measurements have not yet been made to confirm the large edge velocity shear as predicted, e.g., in Fig. 4.

i. Spin-up. While edge rotational flows result from electrical shorting, as just described, dissipation-related effects may participate in the transient “spin-up” of the core plasma. The oldest such is *particle-loss spin-up*;¹¹ ions escaping through the separatrix have a preferential rotational drift. The end loss of these ions produces a rocket effect, causing the FRC to rotate in the opposite direction. Another is *viscous transfer*;¹⁰⁸ edge rotation creates a viscous friction on the core plasma, causing it to spin up. The torque exerted by twisted field lines was sufficient to supply the angular momentum increase. A subsequent comparison with experiment showed that both the particle-loss and viscous mechanisms contribute to spin-up but that the later dominates in most cases and increasingly so for larger FRCs.¹¹⁶ Later, it was suggested that the self-consistent electric fields are such that particle-loss spin-up may, in fact, predict spin-up in the wrong direction.¹⁰⁹ Recently, another possibility has appeared based on the canonical angular momentum P_θ , a constant of motion for individual ions. If P_θ is conserved, the dissipation of poloidal flux leads to an increase in the global angular momentum.^{117,118} This requires a selective dissipation with resistive drag on the electrons (dissipating flux) without an appreciable mutual drag on the ions. A kinetic simulation lends support to this hypothesis.¹¹²

j. Scrape-off layer and jet. A relatively thin SOL (thickness L_n) compared to R_s is an important feature of in elongated FRCs, with $L_n/\rho_{i0} \sim 2 - 4$. Early experimental inferences of L_n are summarized in Ref. 119, where it was noted that the SOL is anomalously broad. Since then several experiments have inferred L_n using multi-chord interferometry: FRX-C/LSM,⁸⁰ PIACE,⁸² FIX,¹²⁰ and FRX-L.^{27,93} More recently, observation of the bremsstrahlung emission radial profile on NUCTE-III led to conflicting results.^{47,81} Reference 84 gives a compendium of L_n inferences. The reason

for the very thin SOL is associated with rapid end loss of unconfined ions. This topic is taken up in Sec. V.

III. GLOBAL STABILITY

A. Introduction

Plasma stability falls into two classes, global or “macro” and micro stability. The former involves disturbances on large scales ($L > \rho_i$, c/ω_{pi}) and low frequencies ($\omega < \omega_{ci}$). Each class poses its own threat: global *instability* threatens the disruption or massive restructuring of the plasma, whereas micro*instability* merely raises the transport rate. This section focuses on the former. Section V will address microstability and transport.

Concern about global stability has dominated FRC research from the beginning and continues to this day, especially regarding one particular mode, tilting. Three principal global instabilities of FRCs have emerged over the years. Figure 5 portrays these in the order of their discovery. The three are fundamentally different. (1) *Rotational* instabilities are ideal modes in that frictional forces play no part. They are driven by the centrifugal (“gravitational”) effect of rotational flow. Potentially *disruptive*, their nonlinear growth can cause the plasma to scrape against the wall, causing catastrophic loss of energy. (2) *Tearing* instabilities are resistive modes, i.e., friction of some kind is a primary player. They cause topological changes in the magnetic configuration. (3) *Tilting* is an ideal mode, driven by “bad curvature.” It is the best known and most feared of the many ideal modes. It is potentially disruptive in its nonlinear growth, which can distort and flatten the X-points, causing rapid reconnection and the catastrophic opening of closed magnetic surfaces.

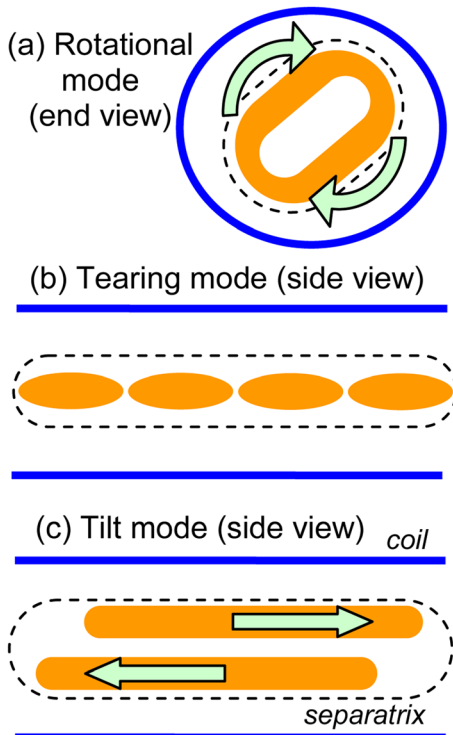


FIG. 5. (Color online) Global instabilities of FRCs.

These three instabilities are cousins to analogous modes in toroidal confinement systems, but with important differences. (1) *Rotational* modes have received relatively little attention in toroidal systems, although the rapid rotation rates of neutral-beam-driven tokamaks may destabilize centrifugally driven modes. (2) *Tearing* in typical toroidal systems occurs at mode-rational surfaces, where the safety factor is a rational number $q = m/n$ (ratio of poloidal to toroidal mode number). However, the classic FRC lacks a toroidal field so that $q = 0$. Thus, tearing is fundamentally global and not associated with isolated surfaces. (3) *Tilting* is analogous to the lowest-order kink mode ($m = n = 1$) in toroidal systems, but there the similarity ends, at least in highly elongated FRCs. Its mode structure (Re. Fig. 5) hardly resembles a rigid tilting. Instead, it is almost purely an internal mode. While its linear phase distorts the separatrix very little, its nonlinear growth sets up an antiparallel field structure on the separatrix, leading to reconnection and disruption.

Tearing and rotational modes were the first to be observed in early FRCs, largely because they were obvious even with the primitive diagnostics available at the time. Most study, however, has focused on tilting because it is the most dangerous for the long-term prospects of FRC confinement. Therefore, this review addresses tilting first and in much greater detail.

B. Tilting – experiment

1. Background

Tilting has a straightforward physical basis. A rigid, current-carrying ring immersed in a background magnetic field is unstable to overturning (or tilting) if the ring current generates a field component *opposite* to the background field. The same tendency occurs in a current-carrying plasma even though the “ring” is hardly rigid. In terms of the MHD energy principle, the destabilizing effect in the δW integral is the pressure-gradient term.¹²¹ FRCs stand in marked contrast to low- β toroidal plasmas, where the primary destabilizer is the current term; this term is zero in classic FRC equilibria which have no parallel current. Unfortunately, stabilizing terms in the energy principle are ineffectual in FRCs, e.g., compressibility (plasma and magnetic); tilting is essentially incompressible.

An early paper addressing spheromaks first identified tilting as a threat in compact toroids.¹²² The 1980s produced several analyses of the ideal tilting in FRCs.⁵ These highlighted three important properties. (1) The nominal growth rate is

$$\gamma_{\text{tilt}} \approx 2V_{A0}/L_s \quad (10)$$

where $V_{A0} = B_{\text{ext}}/(\mu_0 m_i n_m)^{1/2}$ is the reference Alfvén speed (Appendix) and $L_s/2$ is the FRC half-length. For the FRC data compendium,⁸⁴ the median growth time $1/\gamma_{\text{tilt}}$ is a mere 2 μs . As an indication of how fast this is, it equals the free-streaming end-loss time for a θ -pinch plasma with only open magnetic surfaces. (2) In *oblate* FRCs, tilting resembles the more-or-less rigid overturning of a current ring and can be stabilized by a close-fitting conducting wall.¹²² (3) In

elongated FRCs, tilting is primarily an *internal mode* during its linear growth, as portrayed in Fig. 5(c).

a. Tilting detection. By 1988 (Ref. 5), tilting and other MHD instabilities had not yet been observed because adequate methods for detecting internal disturbances were unavailable. Theta-pinch-based FRCs were simply too hot (>100 eV) and dense ($>10^{21}\text{m}^{-3}$) for the insertion of probes. On the other hand, since γ_{tilt} is very large, the survival of an FRC for many times $1/\gamma_{\text{mhd}}$ offered *circumstantial* evidence for tilt stability. For the FRC data compendium,⁸⁴ 2/3 of the data have $\gamma_{\text{mhd}}\tau_N$ between 21 and 63, i.e., typical FRCs survive several dozen ideal tilting growth times. Since 1988, however, several non-intrusive diagnostics for tilting detection have emerged. A *Mirnov array* of magnetic probes mounted just behind the vacuum wall¹²³ picks up stray signals from a primarily internal tilting disturbance. Two experiments, FRX-C/LSM (Ref. 124) and LSX,¹²⁵ have applied these to detect tilting, resolving modes with $n=0,1,2,3$, as well as the evenness/oddness of B_θ with respect to the mid-plane. (Tilting is the $n=1/\text{odd}$ component of B_θ .) More recently, Mirnov arrays have been used to detect radial shift²¹ as well as tilting.^{21,126} Tomography has emerged in recent years as an effective, non-intrusive method for detecting tilting and other internal mode dynamics.^{93,127–129}

2. Elongated FRCs

FRCs have long been created in θ -pinch devices, which naturally form elongated FRCs. However, the newer “merging spheromak” formation method (Sec. VI) naturally forms oblate FRCs. Since tilting is quite different in elongated and oblate FRCs, the relevant experiments are reviewed separately.

The peak value of the Mirnov array signal was used as a measure of tilting strength in FRX-C/LSM (Ref. 124) and LSX.¹²⁵ A *quality* index for transport and stability was defined as the *flux anomaly* $\tau_{\text{cla}}/\tau_\phi$, where $\tau_{\text{cla}} \equiv (R_s/4)^2/(\eta_{\text{cla}}/\mu_0)$ is the classical confinement time.¹²⁴ High $\tau_{\text{cla}}/\tau_\phi$ was interpreted as poor quality and a possible marker of gross instability. Unfortunately, these two experiments showed significant differences for unknown reasons that remain unclear. Here, effects in common between them are listed first, followed by differences.

a. Commonalities between FRX-C/LSM and LSX. (a) FRC birthing by the θ -pinch method is extremely violent with Alfvénic speeds in the radial and, subsequently, axial contractions. The term “implosion” has often been used for both. (b) Quality FRCs become increasingly more difficult (or impossible) to achieve with increased S_* . (c) Quality is extremely sensitive to the formation “recipe.” Fig. 14 of Ref. 130 shows that a change of $2\ \mu\text{s}$ in pre-ionization (PI) timing can cause a factor of ~ 10 change in τ_ϕ . This highlights the importance of an optimized formation recipe. (d) Quality *does not* correlate with p_0 and B_0 (Fig. 19 of Ref. 124); poor quality FRCs can be produced even for lower “easier” values of each.

b. Peculiarities of FRX-C/LSM (Ref. 124). (a) Quality *does not* correlate with the symmetry of the PI plasma (Fig.

19(b) of Ref. 124). (b) Large tilt signal (peak amplitude) is the principal factor correlating with poor quality. Correlation with other modes is relatively poor, Re Figs. 20 and 21 of Ref. 124. (c) Quality consistently plummets for $S_*/E > 3$, where S_*/E is a theory-based tilt stability index (described shortly).

c. Peculiarities of LSX (Refs. 98 and 125). (a) LSX allowed greater flexibility in the formation recipe. The primary control in FRX-C/LSM was the pre-ionization by a ring- θ -pinch (θ -PI), especially its timing. LSX used up to four PI methods with independent timing and amplitude: (i) θ -PI, (ii) z-pinch PI, (iii) barrier fields, and (iv) independent end coils (“bubble” or “trigger” coils). (b) Multiple B_θ modes were consistently active, and the tilt signal ($n=1/\text{odd}$) was not dominant. (c) No correlation appeared between quality and peak B_θ signal (Fig. 17 of Ref. 125). Indeed, large amplitudes were more the rule than the exception, regardless of the subsequent quality. (d) *Global asymmetries* during formation (azimuthal, axial) were the primary “birth defects” leading to poor quality. (e) Quality FRCs were produced for S_*/E up to about 5 (well above the theoretical limit ~ 3).¹²⁵ (f) Formation optimization for larger $S_*/E > 5$ could not be attempted because of early termination of the experimental program. The observation of poor quality in this regime was attributed to a lack of optimization, fostering the claim that “a fixed limit for stability/good confinement has yet to be determined.”

3. Oblate FRCs

Since 1988 (Ref. 5), new formation methods have emerged. One such, spheromak merging produces oblate FRCs, $E < 1$ (Re Sec. VI). Since this method typically produces FRCs with lower densities and temperatures than θ -pinch-based methods, it has been possible to insert multi-axis magnetic probes for direct detection of tilting and other global modes. Measurements in all three spheromak-merging experiments confirm that oblate FRCs *do* tilt at a rate comparable to the MHD growth rate, Eq. (10).^{26,28,48,113,131} Despite the fast growth rate in oblate FRCs, Re Eq. (10), passive stabilization measures have proved effective in suppressing some MHD modes ($n=1$ tilt and radial shift). These include a hollow, conducting central column,^{19,48} close-fitting conducting shells on the sides,¹³² strong external field shaping,⁴⁸ and a nearly enclosing cylindrical shell.²⁶ These methods work best for very low elongation ($E < 1/2$), but are ineffective at stabilizing higher- n siblings of the tilting mode.⁴⁸

C. Tilting – ideal MHD theory

1. Elongated FRCs

The MHD theory of tilting has advanced broadly since 1988.⁵ Consider first elongated FRCs, initial-value codes predicted that tilting indeed leads to disruption in its nonlinear phase.^{133,134} Figure 6 shows results from a nonlinear MHD computation.¹³⁴ Starting with a very small velocity perturbation of $V_{A0}/5000$, tilting grows in eight MHD times (Re Eq. 10) to the large magnetic structure distortion in

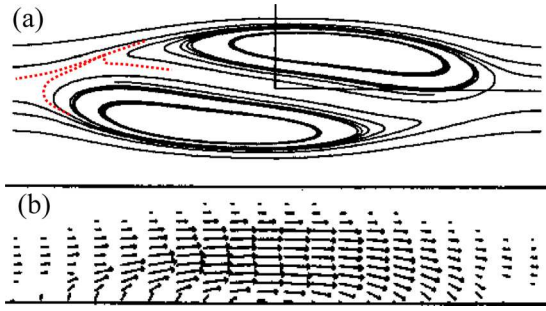


FIG. 6. (Color online) Magnetic field lines and velocity vectors for a tilting FRC (Re Figs. 1,3 of Ref. 134 at $\gamma_{MHD}t=8$). Reprinted with permission from R. D. Milroy, D. C. Barnes, R. C. Bishop, and R. B. Webster, Phys. Fluids B **1**, 1225 (1989). Copyright 1989, American Institute of Physics.

Fig. 6(a). The dotted lines (red) here are an artist's rendition of the X-point distortion ($\psi=0$ magnetic field lines). Observe that the displacement vectors in Fig. 6(b) confirms the earlier reckoning that the tilting mode is primarily internal with only small displacements outside the separatrix. By two growth times later ($\gamma_{tilt}t=10$), the closed field structure is fully disrupted.

a. Profile effects. Early research sought ways to stabilize tilting: flattened pressure profile and modified separatrix shape. These hopes were largely dashed by the mid 1980s.⁵ Later, a simulation predicted that a more racetrack separatrix shape produced by strong end mirrors would reduce the growth rate by a factor of 3–7.¹³⁴ Hopes for complete stabilization for a preferred separatrix or current profile shape revived briefly based on a spectral analysis with a limited set of basis functions,^{41,135} but these collapsed upon application of a more complete basis set.¹³⁶

b. Elongation saturation. A spectral analysis addressed a broad range of equilibria,⁵⁴ considering the effects of *elongation*, *current profile*, and *separatrix shape* and examining modes with $n=1$ and $n>1$ modes with both even and odd symmetry (with respect to the midplane). This showed that the nominal MHD scaling $\gamma_{tilt} \propto 1/E$ (Eq. 10) only held in the isolated case of an elliptical separatrix with a flat current profile. Otherwise, the growth rate *saturated* for $E > 3-3.5$, i.e., its $1/E$ “fall” ceased and $\gamma_{tilt} \sim V_{A0}/3R_s$. This occurs for larger E because the disturbance concentrates in the bad-curvature region near the ends of the FRC. A 3D-MHD initial-value code⁴² confirmed these trends. An analytic treatment of the dispersion relation using a small- $1/E$ expansion¹³⁷ focused on equilibria where $1/E$ scaling still held.

2. Oblate FRCs

A simple but physically insightful general treatment of global modes spanned the range from oblate to prolate FRCs.¹³⁸ Tilt stability in ideal-MHD was predicted for sufficiently oblate FRCs ($E < 1/2$). Rotation and finite ion orbits had little effect. An extensive set of 3D nonlinear MHD simulations⁴³ addressed tilting and its siblings (radial shift mode, intermediate- n “co-interchange” modes, interchange). In oblate FRCs, tilting appeared as an *external* mode, with a

more-or-less rigid tilting of the separatrix envelope; this contrasts sharply with the internalized tilting for elongated FRCs (Figs. 5 and 6). Higher- n modes had higher growth rate $\gamma \sim n\gamma_{tilt}$ (cp Eq. 10). For low $E < 1/2$, the radial shift ($m=0$) replaced tilting ($m=1$) as the dominant $n=1$ mode. The interchange was easily stabilized if β at the separatrix exceeds a relatively small value ($\sim 4\%$). Viscosity reduced the tilting growth rate¹⁰² and may account for the several-times reduction compared to the ideal MHD rate observed in SSX.

An early study of oblate spheromaks¹²² predicted the passive stabilization effect of a close-fitting conducting shell. Nonlinear MHD simulations confirmed the same for oblate FRCs.⁴³ However, shells failed to stabilize higher- n modes. Moreover, finite-Larmor radius (FLR) failed to suppress these modes in hybrid computations (kinetic ions and fluid electrons).¹¹² FLR was too weak to overcome their higher MHD growth rate. Higher- n modes disrupted confinement in 10 or so growth times. An alternate passive stabilizer is a hollow, conducting center column. Nonlinear MHD simulations with line-tying to the column showed growth rate reduction by up to a factor of two, but did not predict stability.⁴⁸

D. Tilting – extended MHD theory

A popular rule of thumb states that “if the plasma is unstable according to ideal MHD theory, then it *will* be unstable in experiment.” Elongated FRCs appear to be an exception to this dictum. This remarkable paradox was the primary focus of the LSX experimental program as well as a long series of theoretical studies searching for the missing stabilization factor. These focused initially on “extended MHD,” corrections to the basic fluid model: flow, two-fluid effects, and gyroviscosity.

Extended-MHD models introduce an intrinsic length scale, something completely missing in ideal-MHD: the ion skin depth c/ω_{pi} in the two-fluid model and the ion Larmor radius ρ_i in the gyroviscous model. While these scales may be small (e.g., $1/10$) compared with global plasma dimensions, they become important in isolated regions of an FRC. The difficulty with extended-MHD analyses is that some elegant properties of ideal MHD are lost: the modes are no longer orthogonal and the mode frequencies become complex. Absolute proofs of stability are lost and less satisfactory criteria must be sought. The earliest extended-MHD model was that of Frieman and Rotenberg,¹³⁹ including equilibrium flows they derived a dispersion relation of the form $T\omega^2 + D\omega + W_{mhd} = 0$, where ω is the mode frequency, T is the inertia, and W_{mhd} is the MHD energy. The only element differing from ideal MHD is the new “middle term” with coefficient D , arising from equilibrium flows. The middle term raises the possibility of a reactive stability. Other extended-MHD effects also led to a middle term. The simple treatment of Ref. 138 addressed two such: rotation and finite ion orbits.

1. Two-fluid effects

Two fluid (2F) effects allow decoupling of ion and electron responses. This introduces an intrinsic length scale, the ion skin depth, or more properly the ion inertial length c/ω_{pi} .^{140,141} The new length scale gives rise to a plasma size

parameter, the ratio of the separatrix radius to the ion inertial length:

$$S_* \equiv R_s / (c / \omega_{pi}). \quad (11)$$

In terms of stability, the primary 2F influence is the Hall effect. It allows the electron (and magnetic) and ion displacements to get out of phase, a result of finite ion mass. A large enough phase difference leads to “reactive” stability. Using a simple trial function approach, the dispersion relation becomes¹⁴⁰

$$\omega^2 + C_{2F} \Omega_* \omega + \gamma_{mhd}^2 = 0, \quad (12)$$

where Ω_* is the relative drift frequency (ions – electrons) and C_{2F} is a coefficient associated with two-fluid effects. For a typical FRC with $\Omega_* \approx 4kT_i / eB_e R_s^2$ and with pressure balance $n_m kT_i = B_e^2 / 2\mu_0$, this leads to the stability criterion

$$S_* / E \leq C_{2F} \approx 1. \quad (13)$$

The trial function approach predicts $C_{2F} \approx 1$.¹⁴⁰ A more recent analysis based on a uniform-elongation approximation for equilibria^{50,51} led to a more optimistic $S_*/E < 2.6 - 4$.¹³⁷

Computations with 2F effects present a more complex picture. A 2F initial value code including the Hall effect,¹³⁴ showed that for high S_* , the tilting growth follows the simple prediction¹⁴⁰ until γ falls to about $(1/2 - 2/3)\gamma_{MHD}$. For lower S_* , γ stops falling and actually begins to rise, a result of a change in the mode structure. The disturbance becomes more complex with the more radial and axial structure, allowing tilting to continue growing and defeating the promising stabilizing tendency. Later studies^{40,44} addressed other equilibria: more hollow current profile, higher separatrix β , and sharply peaked current at the separatrix. The conclusions remained much the same.

Since S_* and E are readily inferred in experiments, the criterion (Eq. 13) can be compared with the FRC data compendium,⁸⁴ as shown in Fig. 7. All the data are for stable, long-lived FRCs, which is at least circumstantial evidence of tilting stability. The “2F” line is the criterion from Eq. (13); points below this line are predicted tilting-stable and points above unstable. Evidently, the simplest 2F theory¹⁴⁰ fails to account for the stability of most of the experiments. Espe-

cially, intriguing is the TRX-2 example with $S_*/E = 7.5$. This result achieved in hydrogen (rather than deuterium) is not an isolated “super shot” but represents a series of experiments.¹⁴²

2. FLR effects–gyroviscosity

Ion kinetic effects, loosely called “FLR,” is the most widely accepted explanation for tilting stability of elongated FRCs. The large ion Larmor radius ρ_i in typical FRCs “stiffens” the plasma against the distorting effect of internal tilting. FLR is a nondissipative effect because it does not rely on frictional collisions. It is strongest near the O-point and X-points where ρ_i becomes large. It is also quite important at the separatrix, where the density gradient length L_n is comparable to ρ_i . The effect of FLR is that the ion displacement gets out of phase with that for the magnetic disturbance, just as for 2F effects. If the phase difference is large enough, tilting is *reactively* stabilized. While both FLR and 2F effects arise from ion inertia, the former results strictly from the ion orbit size.

Several indices have been used to measure the importance of FLR. The oldest is the parameter $S \equiv R_0 / \rho_{i0}$, where ρ_{i0} is the ion Larmor radius based on the external field (Appendix). A better index recognizes that the local ρ_i in the interior can be much larger than ρ_{i0} : $\bar{s}$ is the ratio of the minor radius to an *averaged* ion Larmor radius

$$\bar{s} = \int_{R_0}^{R_s} r dr / R_s \rho_i. \quad (14)$$

The difficulty is that $\bar{s}$ requires knowledge of the internal magnetic structure, which is almost never measured directly, forcing one to rely on theoretical estimates of the internal structure. For “rigid-rotor” profiles subject to the axial force balance relation, $\bar{s} \approx SX_s/5$. Because of this uncertainty, the 2F index S_* (Eq. 11) is a more practical index because it depends on quantities (R_s , average density) that are routinely measured. Actually, for high- β plasmas, S_* is a reasonable measure of FLR since the two length scales are closely related: $\rho_{i0} / (c / \omega_{pi}) = (\beta_i / 2)^{1/2}$, where $\beta_i \equiv 2\mu_0 n_m kT_i / B_e^2$ is the maximum ion β based on the external magnetic field. For $T_e = T_i$, $\beta_i \sim 1/2$ so that $S/S_* \sim 2$.

The original FLR stabilization criterion dates to the early 1960s,¹⁴³ $\gamma_{mhd} / \omega_{ci} \leq (k\rho_i)^2$, although most researchers have used the less stringent condition that the growth rate be smaller than the diamagnetic drift frequency (also regarded as an FLR effect):

$$\gamma_{mhd} / \omega_{ci} \leq k\rho_i^2 / L_n, \quad (15)$$

where k is the wave number of the mode and L_n is the density scale length. Apply this to tilting by taking $k = n/r$, $n = 1$, $r = R_s / \sqrt{2}$, γ_{mhd} from Eq. (10) and $\omega_{ci} = eB_e / m_i$, leads to $S_*/E < \beta_i \sim 1/2$ for FLR stability, i.e., FLR also gives rise to an S_*/E stability criterion. Early attempts to apply FLR to FRCs failed because the FLR expansion fails at magnetic field nulls where ρ_i is singular.⁵ However, gyroviscosity,⁵² the fluid-based representation of FLR, offers a non-arbitrary way around this difficulty: the gyroviscous force vanishes as

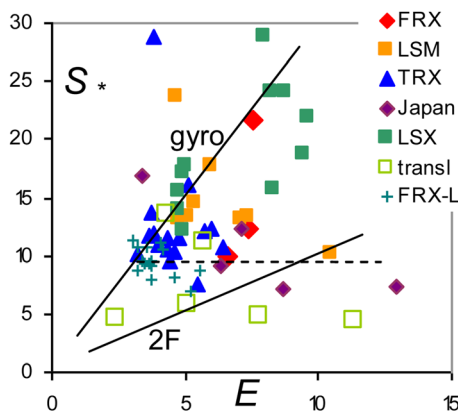


FIG. 7. (Color online) Size parameter vs separatrix elongation for elongated FRC's.

the ions demagnetize (near field nulls), avoiding a singularity.¹⁴⁴ Variational forms were derived for gyroviscous plasma,¹⁴⁴ one including the parallel viscosity as well.¹⁴⁵ This gives rise to a dispersion relation as Eq. (12) except with a gyroviscous coefficient C_{gyro} instead of C_{2F} :

$$S_*/E \leq C_{gyro}. \quad (16)$$

Various studies found slightly different values of C_{gyro} . A trial function approach found $S_*/E < 1.6$.¹⁴⁴ More realistic equilibria and a small basis set led to $S_*/E < 3.5$ (Ref. 146) and $S_*/E < 3$ for a fairly broad basis set.¹³⁶ A “rigid” displacement paradigm found $S_*/E < 2.2$ and with the inclusion of rigid rotation $S_*/E < 2.8$.¹³⁸ The line marked “gyro” in Fig. 7 corresponds to the $S_*/E < 3$ criterion. Most of the data lie below this line, i.e., in the tilting-stable region.

The “elongation saturation” effect noted earlier calls this into question. For all, but a limited class of equilibria, the S_*/E scaling breaks off above about $E > 3$, where the criterion becomes roughly $S_* < 9$.¹³⁶ If so, then the predicted stable region is much smaller. Only a small fraction of the data in Fig. 7 lie in the stable region below both the $S_*/E < 3$ and $S_* < 9$ lines. The explanation may be that FRCs naturally have an elliptic separatrix, a condition for the “uniform-elongation” approximation^{50,51} for which the S_*/E criterion remains valid for all elongations. Limited studies of equilibrium studies^{46,85} suggest that more elongated FRCs naturally assume a racetrack-like shape.

Finally, even the prediction of absolute tilting stability fails. A spectral study with a much more complete basis set showed that below a supposed critical value of S_*/E (or of S_*), the growth rate merely decreased but showed no tendency to go to zero, Re. Fig. 2 of Ref. 55. This agrees with an early no-absolute-stability finding using fully-kinetic ions,¹⁴⁷ as well as a modern kinetic treatment to be discussed shortly.⁴⁵

E. Tilting – kinetic theory

By 1988 (Ref. 5), kinetic models of global modes in FRCs were in their infancy. With one exception¹⁴⁸ these were limited to analytic or spectral analyses. Since then the explosive growth of supercomputing power has fostered increasingly ambitious studies, allowing the use of full-fledged initial value simulations.

The kinetic models investigated to date are as follows: (1) A *MHD-Vlasov hybrid* model combined a background of fluid ions and massless electrons (Hall MHD) with a kinetic “beam ion” population (collisionless Vlasov species).⁷⁰ (2) A *particle-in-cell* (PIC) treatment received a limited early treatment,¹⁴⁹ but since 2000 has become a broadly useful tool. While two studies considered a small set of equilibria with modest elongation ($E \approx 1, 2$),^{101,150} extensive studies by Belova and colleagues addressed a wide range of equilibria from oblate to highly elongated.^{42–45,112} These featured the *δf method*, which though limited to examples with analytic forms of the equilibrium ion distribution have much lower noise than standard PIC simulations, at least in the linear growth phase. (3) The *full kinetic treatment* is the most ambitious. An early study carried out a direct integration of ion orbits in a background of adiabatic fluid electrons but

was limited to the linear growth phase.¹⁴⁸ A later simulation treated both ions and electrons kinetically, albeit with an artificially small ion-to-electron mass ratio of 50.^{96,151}

a. Linear mode growth. Kinetic models have generally confirmed gyroviscous predictions of the linear growth rate, although these were limited to equilibria with nearly elliptical separatrix shapes. Kinetic effects reduce the linear growth

$$\gamma \approx \gamma_{mhd} \exp(-3E/S_*), \quad (17)$$

where γ_{mhd} is the nominal MHD rate in Eq. (10). Figure 8 shows results from one of the studies. The dashed line, Eq. (17), holds for E/S_* less than about 1/2. These studies also addressed particular physics effects. One isolated FLR from Hall effects by turning off the Hall term in the Ohm’s law.⁴⁴ The Hall effect significantly modified the mode structure (making it more complex as in Ref. 134) and giving the mode a substantial real frequency, the Hall effect only modestly reduced the FLR growth rate. Finite electron pressure increased the tilt growth rate, a result of an effective reduction of ion FLR effect.⁴⁴

A marked exception to these trends appeared in the standard PIC simulations of Ref. 101. These began with an equilibrium with a moderate “up-down” toroidal field on opposite sides of the midplane (zero net toroidal flux) as well as poloidal ion flows comparable to the Alfvén speed. Studies limiting the disturbance to $n=0$ and 1 components showed no clear instability, even for the largest FRCs considered ($S_*/E = 16$). This striking difference from the *δf* studies⁴⁴ may arise from different equilibria: those in the *δf* studies had no poloidal ion flow and negligible toroidal field. Whether the higher noise level in the standard-PIC simulations had a significant (but nonphysical) effect is unclear. Subsequent standard-PIC studies,¹⁵⁰ which also allowed higher- n components ($n=2,3,4$), exhibited a multi-mode instability that caused a major restructuring of the equilibrium but not disruption. This possibility may reflect a relaxation event, as addressed in Sec. IV.

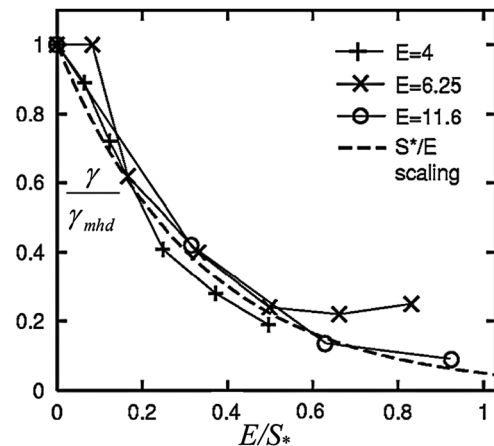


FIG. 8. FLR stabilization effect on tilt mode, Re Fig. 1 of Ref. 112. Reprinted with permission from V. Belova, R. C. Davidson, H. Ji, and M. Yamada, Phys. Plasmas **13**, 056115 (2006). Copyright 2006, American Institute of Physics.

b. No absolute linear stability. Closer scrutiny of Fig. 8 shows that the growth rate does not generally continue to fall exponentially but levels off below $S_*/E < 2$ (above $E/S_* > 1/2$ in the figure) at 1/10 to 1/4 of the MHD rate. This is not “elongation saturation” since the examples in Fig. 8 are normalized to γ/γ_{mhd} . Further studies showed that the leveling-off of γ/γ_{mhd} arose from resonant ion effects. Resonant ions support a residual tilting instability even in highly kinetic FRCs.^{42,44}

c. Profile effects. A fully kinetic treatment (kinetic ions and electrons) addressed how the equilibrium affects linear growth of tilting.¹⁵¹ While these studies considered realistic values of S_*/E (2.5 – 6.5), they were limited to equilibria with much higher X_s (0.7 – 0.85) and lower elongation ($E < 2.5$) than in experiments. As a consequence, the separatrix β was somewhat lower than in experiments. Raising β_{sep} from 0.02 to 0.2 stabilized tilting (or reduced its growth rate by a large factor ~ 30), an effect ascribed to the stabilizing influence of ions spending a large fraction of their orbits outside the separatrix in the region of “good” curvature. However, these simulations were only run for very short times (up to 4 Alfvén times); consequently, growth may not yet have converged to the accurate linear rate. Later studies⁹⁶ showed that both hollow current profile and high β_{sep} reduce the tilting growth rate; although the reductions were much less impressive (e.g., factors of 5). Clearly, differences between δf -PIC and the kinetic electron-ion simulations remain unresolved: in particular, the observation in the latter that S_* has no consistent effect on stability and in one case higher S_* actually improves tilting stability.

F. Tilting – other effects

1. Nonlinear saturation

Nonlinear MHD simulations of tilting were unanimous: instead of a nonlinear saturation, tilting grew until complete disruption.^{133,134} In this regard, note the extreme distortion of the X-point produced by the tilting disturbance shown in Fig. 6 (red dashed lines), setting up a thin anti-parallel field structure leading to rapid opening of internal flux surfaces; in short, disruption. However, kinetic studies have shown that disruption is not always the nonlinear consequence. A δf -PIC simulation of a case with $S_*/E \sim 3.2$ showed nonlinear saturation of tilting, Re Fig. 9.⁴⁵ The $n = 1$ disturbance saturates and begins to decrease at $t/\tau_A \sim 60$, i.e., in about 8 linear growth times in this example (Re Fig. 4 of Ref. 45). The FRC at these times shown is not axisymmetric, mainly because the $n = 2$ rotational mode succeeds tilting as the primary disturbance. On the other hand, a simulation with $S_*/E \sim 15$ shows rapid dissolution of the configuration in 15 – 20 τ_A (7 – 10 linear growth times).¹⁰² Here, the kinetic effect is weak enough that it only reduces the MHD growth rate by about 10%. Nonlinear stability of the $n = 1$ mode in oblate FRCs has also been observed in an MHD simulation, although a large viscosity may have figured in the late-time suppression of the mode.

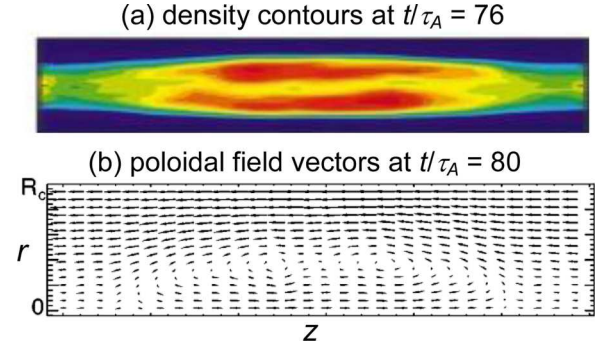


FIG. 9. (Color online) Nonlinear tilting saturation: (a) excerpt from Fig. 5; (b) excerpt from Fig. 6 of Ref. 45, where $\tau_A \equiv R_c/V_{A0}$. Reprinted with permission from E. V. Belova, R. C. Davidson, H. Ji, and M. Yamada, Phys. Plasmas **11**, 2523 (2004). Copyright 2004, American Institute of Physics.

Three possible reasons for tilting saturation in moderate ranges of S_*/E come to mind. (1) The instability growth is *multi-mode* and higher- n modes grow precipitously (especially $n = 2 - 4$) simultaneously with tilting saturation, as observed in LSX.¹²⁵ Mode-mode coupling may drain energy out of the tilting mode as fast as MHD instability feeds it. (2) The saturation ($t/\tau_A \sim 50 - 60$ in Fig. 9) may dissipate some of the internal magnetic flux, effectively reducing S_*/E to a lower, “safer” range, although details supporting this possibility were not presented in Ref. 45. (3) The X-point region remains more-or-less intact, at least on visual inspection of Fig. 9, contrasting sharply with its extreme distortion in the MHD simulation (Fig. 6(a)). Ion kinetic effects may stiffen the X-point region. Recall that the tearing open of closed magnetic surfaces necessarily occurs at the X-points.

Consider these observations in view of the FRC data compendium.⁸⁴ Figure 10 shows $\gamma\tau_N$, as a function of the empirical stability parameter S_*/E , where γ is taken from Eqs. (3)–(8) and τ_N is the particle lifetime. In effect, $\gamma\tau_N$ is how many tilt growth times that the FRC survives. Also shown as a heavy dark circle is $\gamma t = 8$, the time to tilting in the computed $S_*/E \approx 3.2$ example (Fig. 9). Comparable FRC experiments have lifetimes a factor of two or more longer than this saturation time, i.e., they easily live long enough to reach a possible saturation. Moreover, the highly dynamical formation phase is likely to create initial tilting components

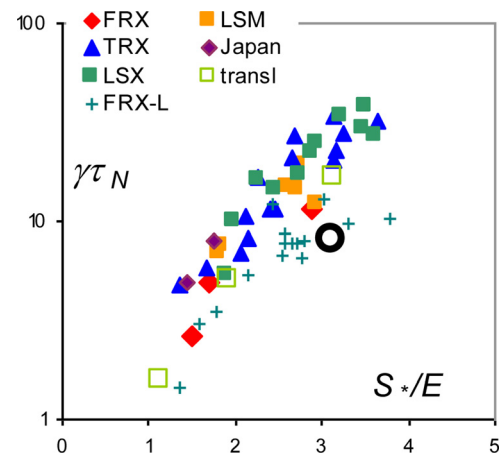


FIG. 10. (Color online) Number of kinetic tilting growth times FRC particle lifetime

larger than the $0.01V_A$ or smaller initialization typically in computations.

2. Flow-shear stabilization

a. Rigid rotation. An early analysis¹³⁹ included the effect of equilibrium flow on stability. MHD simulations addressed the possibility of flow stabilization by rigid rotation.¹³⁴ The primary effect of the rotation was to greatly alter the tilting mode structure, forcing the displacement vectors to be almost purely axial. The mode became an *external* mode with significant distortions of the separatrix, and in so doing defeated any stabilizing tendency of rotation. This might be overcome by imposing strong mirrors at the ends, which substantially reduces the growth rate.¹³⁴ A simple analytical treatment assuming rotation at the ion diamagnetic drift speed predicted rotational stability for $S_*/E < 1.7$.¹³⁸

b. Flow shear. A more promising effect is flow shear, which can shred a nascent MHD mode. Two brief studies investigated this possibility in elongated FRCs. An MHD study¹³⁴ introduced a specified difference in rotational flow of $\Delta u_\theta = 3V_{A0}/4u_\theta$ between the O-point and separatrix. This reduced the growth rate by a factor of three. In a PIC study, flow shear dramatically altered the mode structure, forcing it to localize in regions of lower shear,⁴² the best stabilization occurred for flow shear concentrated near the separatrix. Flow resulting from turbulence-produced broken magnetic surfaces (Sec. II) creates a natural flow shear concentrated near the separatrix.⁸³ However, although flow shear reduced the growth rate, it did not lead to stability.⁴² A PIC simulation considered oblate FRCs,⁴³ even a very large rotational flow difference between the O-point and separatrix only produced a modest reduction in the growth rate.

The foregoing were fairly limited studies of flow-shear stabilization. For example, they did not reference the rule-of-thumb for flow shear stabilization, $|V'| > \gamma$, where V' is the flow shear and γ is the growth rate of the mode in question.¹⁵² End-loss outside the separatrix causes quite steep gradients near the separatrix, with length scale L_n comparable to the ion gyroradius ρ_i . The resulting diamagnetic drift V_* produces a local velocity shear $V' \sim V_*/L_n$. Since $V_* \sim kT_i/eB_e L_n$. It follows, using the MHD growth rate, Eq. (10), that $|V'| \sim (ER_s/\rho_i)\gamma$. Thus, the rule-of-thumb criterion is satisfied by a wide margin (a factor of roughly ES_*) in the separatrix region. The flow shear will be much lower in the interior where tilting may still grow. The flow shear may stiffen a layer of plasma near the separatrix and force the tilting disturbance inward, a tendency suggested by a PIC simulation.⁴²

3. Other physics

Another proposed effect not yet studied in detail is a trapped-electron effect, the so-called “charge uncovering effect.”¹⁵³ Electron trapped in the magnetic well regions near the X-points leads to in a fast rotational drift motion. Since its frequency is large compared to the tilting mode frequency, these electrons will not respond in a fluid-like manner to the tilting disturbance. This is likely to stiffen the otherwise vul-

nerable X-point region and force the tilting mode inward. Modelling this requires a kinetic treatment of the electrons, which has been done, albeit with an artificially small mass ratio m_i/m_e .^{96,151} The effectiveness of trapped electrons in improving stability relies on there being a significant fraction of fast rotating electrons; whether this is the case is unclear. Finally, an effect meriting further study is spontaneous relaxation toward a naturally stable state. This is addressed briefly in the context of tearing stability in Sec. IV.

4. Local modes

Interchange is a traditional characterization of local modes. It is stabilized fairly easily if the separatrix pressure is not too low.^{43,86} *Cointerchange* or *ballooning* modes may be more difficult because they concentrate in regions of bad curvature and can have somewhat higher growth rates than tilting, e.g., $\gamma \sim V_{A0}/R_s$ (cp Eq. (10)). Note that in FRCs, ballooning modes have high- n number, whereas in tokamaks they have high- m ; this difference arises because the dominant magnetic field in FRCs is poloidal rather than toroidal. Ballooning was addressed in early studies reported in the 1988 review⁵ and has also been considered more recently.

In *elongated FRCs*, MHD modeling of $n > 1$ modes confirmed that they have a relatively high growth rate,¹³⁴ e.g., the $n = 5$ mode grows four times faster than the tilting mode. Spectral treatments based on a local-MHD treatment led to a similar conclusion.^{154,155} However, this trend may be equilibrium dependent. Further, because of the finer scale of higher- n modes, FLR (gyroviscous) effects should be moderately effective in stabilizing them, as indicated by both spectral⁵⁵ and PIC (Ref. 45) treatments. The simple FLR stabilization criterion (Eq. (17)) can be applied to ballooning. For $\gamma_{mhd} = V_{A0}/R_s$, and $k = n\sqrt{2}/R_s$, they are stable roughly for $n \geq S_*/4$. Thus, $n \approx 2 - 5$ should be unstable in typical FRCs. Co-interchange may figure in FRC relaxation events; a PIC simulation (Fig. 4 of Ref. 44) shows them exploding into prominence during the nonlinear saturation of tilting.

In *oblate FRCs*, however, co-interchange modes are significantly more virile than tilting and radial shift modes. They are not suppressed by close-fitting conducting walls and flow shear.⁴³ FLR is ineffective against the large MHD growth rate. Moreover, in their nonlinear phase, they cause disruption of the configuration.⁴⁸

5. Summary of MHD stability

In *oblate FRCs*, global modes can be stabilized by close-fitting conducting shells, a center column or strong external field shaping.⁴⁸ However, local modes are more difficult and will require active stabilization methods (see shortly).

In *elongated FRCs*, the global stability remains confusing. The apparent empirical limit on S_*/E (Re Fig. 7) conforms with simple treatments of tilting but fails with more complete models. The question remains whether the so-called $S_*/E \leq 3$ limit represents a fundamental instability limit¹²⁴ or a limitation imposed by sensitivities of FRC

formation in a θ -pinch.¹²⁵ Particularly, intriguing are the high S_*/E (~ 7) results in hydrogen, which strongly violates the usual limit without suffering compromised confinement.¹⁴² Several factors that may suppress global instability have received only cursory treatment. These include sheared flows and trapped electron effects (both of which can push the disturbance away from the vulnerable X-points), and plasma relaxation toward a preferred equilibrium with stabilizing flows or toroidal fields.

Lacking a convincing picture of global stability, a “quality of confinement” index, as proposed in Ref. 124, remains the best measure of stability. The quality yardstick used there was the confinement time based on classical resistivity. However, it is well known that the decay of magnetic flux and particle inventory is anomalous and the driving mechanism unidentified (Re Sec. V). A better standard measure of confinement is the widely used empirical scaling⁷⁵ R_0^2/ρ_{i0} . Moreover, τ_N rather than τ_ϕ should be used for the confinement time since the disruptive effect of tilting (if any) occurs at the separatrix and causes loss both of particles and flux. Figure 11 shows the quality index $R_0^2/\rho_{i0}\tau_N$, lower values mean higher quality. The slightly tilted solid line is a “best quality” empirical line. The vertical dashed line at $S_*/E \approx 3.7$ corresponds roughly to the familiar empirical S_*/E “stability” limit (Re Fig. 7) above which FRCs are either tilt unstable or difficult to form. Curiously, the hydrogen example from TRX-2 (Ref. 142) lies along the “best quality” line but widely violates the S_*/E “limit.”

G. Rotational and tearing instability

1. Rotational mode

The first destructive instability observed in experiments was the $n=2$ rotational mode (Fig. 5), driven by the centrifugal “gravity” effect of rotation. It is readily detectable by end-on framing photography and by the appearance of an oscillating signal in single-chord interferometry. Its effect in the low-density halo region has recently been detected using electric field probes.¹⁵⁶

The rotational mode limited the lifetime of essentially all early experiments. Its remarkable disappearance in experiments in the Soviet Union was what motivated the re-emergence of FRC research in both Japan and the United States.

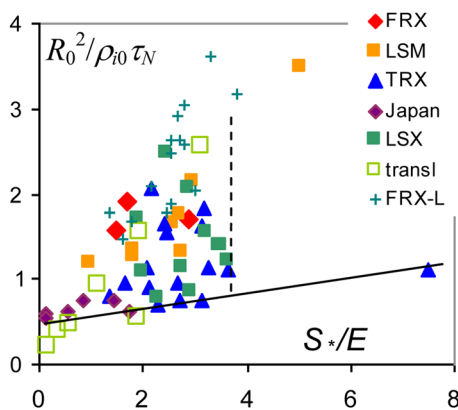


FIG. 11. (Color online) Quality of confinement vs MHD stability index.

Later, active stabilization of the rotational mode by simple multipole windings was demonstrated in the early 1980s (as will be discussed in Sec. III D). Even so, the rotational mode is occasionally revisited. MHD (Ref. 134) and kinetic^{43,45} computations consistently exhibit the rotational mode, although it is almost always overshadowed by tilting. Mirnov-probe measurements show that the rotational mode appears early, during startup, alongside tilting ($n=1$), but then decays; it then reappears relatively late in time, leading to disruption.^{124,125} In view of its late appearance and the proven ability to suppress it by multipoles, the rotational mode is generally regarded with less concern than tilting. A modest rotation-driven “wobble” mode ($n=1$) is also observed in some experiments. While its effect is benign, it is a nuisance for the interpretation of diagnostics that rely on a well-centered FRC.

Several questions remain about the rotational mode. (1) FRCs formed by the “translation” method (Re Sec. VI) appear to be immune to the rotational mode, even at late times.^{103,157,158} A simple analytic theory in parallel with two-fluid computations¹⁵⁹ suggest that this stability may result from the modest toroidal field that appears naturally with this formation method.¹⁰³ (2) In FRCs formed and sustained by rotating-magnetic fields (RMF), the RMF stabilizes the rotational mode if the separatrix is fairly close to the wall. This has been ascribed to the “surface” pressure effect of RMF.⁹⁰ (3) Rotation is the driver for the rotational instability, but the cause of rotation or “spin-up” remains unclear, although several explanations have been proposed, as discussed in Sec. II.

2. Tearing

Disruptive tearing, as shown in Fig. 5, was clearly observed in several FRC experiments in the 1960s but was controlled by improved purity and external field uniformity. Since then, tearing instability in the FRC core has not been a threat. The issue of tearing in the immediate FRC context was addressed once theoretically, but it is suppressed if R_s/R_c is not too small, i.e., wall stabilization.¹⁶⁰ However, a recent study showed indirect evidence that tearing instability acts as a *benign* agent to regulate the FRC internal profile.¹⁶¹ If so, its primary effect is to enhance transport and as such it is addressed in Sec. V.

H. Active stabilization methods

a. Multipole fields. As described earlier, *passive* conducting shells and a center column have proved effective in suppressing global modes in oblate FRCs (Sec. III). Such measures are ineffective against internal modes in elongated FRCs. On the other hand, other measures (modest multipole fields and RMF) have proved effective in suppressing the rotational mode in elongated FRCs. Theory gave a reasonable explanation for this.^{5,162} The predicted marginal stability condition corresponds to a balance between the restoring force of the multipole fields acting “on the surface” of the FRC and the destabilizing centrifugal force.¹⁶² Multipoles have not always been used in experiments, mainly because rotational instability often appears late in the natural decay lifetime.

Even so, they have occasionally been investigated.^{16,163,164} Besides stabilizing or delaying¹⁶ the $n=2$ mode, they also tend to suppress the wobble and keep the FRC well centered.^{163,164}

b. Beam ion component. Another active stabilization technique was proposed long ago for suppressing global modes⁵ a rapidly rotating “beam-ion” component, which would have a stiffening effect on the plasma. This might be provided by ion-ring injection, as proposed for a spheromak¹⁶⁵ or neutral beam injection (NBI). NBI has been reported in a single facility, FIX.^{166,167} A modest neutral beam (14 keV energy, 23 Amps, 10 ms pulse) was injected at an oblique angle into FIX. The beam suppressed the wobble ($m=0$, $n=1$), a common effect in “translation” formation.

A hybrid model addressed the stabilizing effect of “beam ions” in elongated FRCs with fluids ions and electrons plus a rapidly rotating kinetic (“Vlasov”) ion component.⁷⁰ Energetic “beam ions” were predicted to be an effective stabilizing mechanism in both prolate and oblate FRCs, provided they carry a significant fraction of the total plasma current. For example, with rapid beam rotation speed ($> 10V_{A0}$), stabilization was predicted if the beam energy exceeded about 40% of the total (beam + plasma) energy. This was achieved despite the fact that the beam ions only accounted for a small fraction ($\sim 2\%$) of the total ion density. A later δf -PIC hybrid study (kinetic ions and fluid electrons) addressed oblate FRCs.¹¹² The combination of close-fitting conducting wall and rapid beam-ion rotation ($\sim 6V_{A0}$) and modest beam density ($\sim 3\%$ of the background ions) stabilized tilting and radial shift even for a fairly large $S^* \sim 18$. The growth of higher- n modes ($n=3,4$) was reduced as well and they appeared to saturate at the time that the $n=1$ disturbance peaked and began to decay.

Rotating magnetic fields (RMF), which are used as a current drive technique (Sec. VII), appear to have a stabilizing effect on rotational⁹⁰ and, in combination with external field shaping, on other instabilities.¹²⁶

IV. SELF-ORGANIZATION

A. Introduction

Self-organization (SO) is traditionally associated with certain magnetic configurations, reversed-field pinches (RFP), and spheromaks. The word has not traditionally appeared in the same sentence as “FRC.” One of the surprises in recent years has been the emergence of the possibility that FRCs may indeed be “natural states,” the product of SO. It is important to keep in mind the definition: SO, or “relaxation” as it is often called, may involve a chaotic *but not catastrophic* transition; it is a kind of *perestroika*, restructuring the plasma but not causing its dissolution.

What catapulted the idea of SO onto center stage of plasma physics was Taylor’s celebrated discovery in 1974 (Ref. 168) of spontaneous relaxation in reversed-field pinches (RFPs). In treating the quiescent phase of a RFP, without specifically invoking MHD processes, he was able to predict the quiescent phase of an RFP by postulating it to be a minimum-energy state subject to conserved magnetic hel-

icity. These “Taylor states,” as popularly called, are force-free with uniform pressure. Studies as early as the 1950s had investigated constrained minimum-energy states, but these “mathematical curiosities” gained no traction in the plasma community.^{169–172}

After Taylor’s remarkable discovery, attempts followed to apply it more broadly than to just RFPs. With the notable exception of spheromaks, this was largely unsuccessful. Tokamaks, e.g., did not form Taylor states. Further, RFPs and spheromaks often have $\langle \beta \rangle \sim 10\%$, whereas a Taylor state is force-free ($\beta=0$). Others attempted to find an alternative or more general SO principle, a generalization that might apply to a larger class of plasma systems and allow finite- β .^{173–176} The subsequent evolution of the RFP concept has moved away from Taylor relaxation as its centerpiece because SO processes compromise energy confinement. Even so, SO has persisted as a topic of high interest: magnetic helicity injection offers a way to sustain a spheromak or “start up” a spherical tokamak. Moreover, it may occur naturally in space plasmas. Several reviews have discussed SO in detail.^{177,178} Finally, SO has emerged of late as a possible controlling process in FRCs.

Two clues encouraged the prospect that FRCs may relax to SO states. (1) Given the *shear violence of the birthing process*, it is remarkable that FRCs can “pull together” to form a long-lived, quiescent state. The Alfvénic and super-Alfvénic speeds and chaos of formation give the plasma every excuse to go unstable and self-destruct. (2) *Profile consistency*: a particular pattern appears in the current density structure. This review considers the two alternative and perhaps complementary methods of finding SO FRC states: a *variational principle* subject to global constraints and a *global tearing stability principle*.

B. Variational principles

The Taylor principle is one of a host of candidate variational principles for finding SO states. These share two common features: (a) a global quantity that is extremized and (b) one or more global constraints. For example, the Taylor principle *minimizes* the magnetic energy subject to a *constrained* magnetic helicity. The credibility of such principles rely on the tendency of the former toward some minimum (or maximum) and the ruggedness of the latter in the face of the turbulent transitional dynamics.

a. Extremized quantity. A host of extrema principles have been proposed over the years. Most studies have focused on *minimum energy*,^{168,172,176,179–181} but others have pursued other principles: *maximum entropy*,^{173,175} *minimum dissipation rate*^{182,183} (a descendent of the earlier “minimum rate of entropy production” principle in non-plasma systems), *minimum rate of change of autocorrelation*,¹⁸⁴ and the *coercive functional* approach which includes a minimization as one of its steps.¹⁸⁵ Most of these fall into two types: (1) ideal-dynamics dominant or (2) friction dominant. These may be distinguished by a Lundquist number (Lu) criterion (ratio of resistive time to Alfvén time), as supported by spheromak simulations¹⁸⁶ but no FRC-relevant computations have been

made. The focus here is on the energy minimization principle, i.e., the high-Lu case. It should be noted that maximized entropy and a minimized energy were shown to be equivalent.¹⁸¹

The immediate question is which energy to minimize. Taylor minimized the global magnetic energy $W_m = \int (B^2/2\mu_0) d^3x$. While this makes some sense in this because the magnetic energy is an “ordered” energy form, it ignores the other ordered form, namely the flow energy. The total ordered energy, the magnetofluid energy, is the sum of the two,¹⁸⁷

$$W_{mf} = \int \left(\sum_j m_j u_j^2 / 2 + B^2 / 2\mu_0 \right) d^3x, \quad (18)$$

which is the energy form minimized in several studies.^{171,172,176,179–181,188–191} The transitional process converts some of the “ordered” energy into “disordered” or thermal energy $W_{th} = \int [p/(\gamma-1)] d^3x$. This is consistent with the type-1 variational principle (ideal-dynamics dominant), where the total energy $W_{mf} + W_{th}$ should be constant in a closed system. This is not the case in type-2 friction-dominated principles, where a continuous energy source is presumed.

c. Constraint. In the absence of a global constraint, the minimum ordered energy is zero, i.e., complete dissolution of the magnetic configuration. A long list of candidate global constraints have appeared, some of which have not been pursued beyond the proposal stage.^{174,175} The more obvious ones are the system mass,^{171,176,181} the system energy ($W_{mf} + W_{th}$),^{176,181} and the angular momentum about a axis of symmetry.^{171,176,181} However, the most attention has focused on various helicities. The invariance of each of these has some basis, i.e., they are global constants of motion in one model or another. These are the kinetic or fluid helicity,¹⁸⁸ $K_f = \int \mathbf{u} \cdot \nabla \times \mathbf{u} d^3x$, the magnetic helicity^{168,169,188,192} $K_m = \int \mathbf{A} \cdot \nabla \times \mathbf{A} d^3x$ (recall that $\mathbf{B} = \nabla \times \mathbf{A}$), the cross helicity^{171,176,188} $K_X = \int \mathbf{u} \cdot \mathbf{B} d^3x$, and the generalized helicity for the ion fluid^{179,190,191,193,194} or for each species “ j ,”^{180,181}

$$K_j = \int \mathbf{P}_j \cdot \nabla \times \mathbf{P}_j d^3x \quad (19)$$

where $\mathbf{P}_j = q_j \mathbf{A} + m_j \mathbf{u}_j$ is the canonical fluid momentum. Note that for massless electrons $K_e \rightarrow K_m$. The global invariance of a particular helicity hinges on the existence of a local “helicity transport equation,” i.e., a continuity principle for the helicity density, e.g., $\mathbf{u} \cdot \nabla \times \mathbf{u}$ for the kinetic helicity. A simple fluid has a transport equation—for K_f .¹⁸⁸ For single-fluid MHD, transport equations have been derived for K_m ^{171,192} and K_X .¹⁸⁸ In fact, the K_f -transport equation also applies to single-fluid MHD,¹⁸¹ although there are reasons (see shortly) why K_f should not be treated as an invariant. In a two-fluid, a K_i -transport equation has been derived,^{179,193} which in fact applies to any species of a multi-fluid (including the electrons).¹⁸⁰ Note that the cross helicity is only a global invariant within the artificial construct of single-fluid MHD; it is *not* an invariant for the more general two-fluid or multi-fluid models.

d. Ruggedness of invariants. In the generally chaotic transition from an initial to a relaxed state, the ideal invariant (e.g., K_m) must be more rugged than the quantity that is minimized (e.g., W_{mf}). Three criteria for this have been proposed: (1) *Inverse cascade*: the direction of the mode-to-mode cascade in the transition must include a branch creating larger-scale structures.¹⁸⁹ (2) *Selective decay*: the helicity in question must dissipate more slowly than W_{mf} .^{173,177} (3) *Stability to tearing*: the helicity in question must be stable to resistive modes, i.e., the time scale for topology change must be shorter than that for helicity change.¹⁹⁵ Applying these tests to a multi-fluid, K_m and K_j pass all three, K_f fails (1 and 2), and K_X fails all three, Re references in Ref. 181. If K_j is indeed a rugged invariant, then equilibrium flows are an essential feature of SO states. Whether this is the case in practice remains a controversial question. Indeed, a very early opinion¹⁷⁰ expressed doubt whether the minimization principle could be generalized to allow nonflowing equilibria. Later, it was claimed that viscosity destroys K_j ,¹⁹⁰ although this was countered by an argument based on the length-scales on which the viscosity coefficient applies.¹⁸¹ Even so, challenges to its durability appear from time to time.¹⁸⁵

e. Euler equations. Variational principles lead to Euler equations for the SO structure. Each species (each K_j constraint) gives rise to an associated Euler equation^{141,180,194}

$$n_j \mathbf{u}_j = \lambda_j \nabla \times \mathbf{P}_j, \quad (20)$$

where λ_j is the associated Lagrange multiplier. If global angular momentum is included as a constraint, an additional term appears on the left side of Eq. (20), $-\Lambda r \hat{\theta}$, where Λ is the Lagrange multiplier associated with the angular momentum.¹⁸¹ The Euler equations are equivalent to the equations for “double-curl Beltrami flows” discussed in Ref. 141. Evidently, the Euler equations govern flows of each species. Note that in the absence of ion flow (electrons carrying all the current), Eq. (20) reduces to the familiar force-free equation $\mathbf{j} = \lambda \mathbf{B}$.^{168,169}

f. “Relaxed” Bernoulli equation. The thermodynamic pressure or temperature is governed by Bernoulli equations, one for each species “ j ” (Re Sec. II). In flowing equilibria, these are $5kT_j/2 + m_j u_j^2/2 + q_j \phi = H_j(P_{j0})$, i.e., the “total enthalpy” is uniform on a drift surface $P_{j0} = \text{const}$. (P_{j0} is the canonical fluid momentum for species j .) In a SO state, however, the function on the right side becomes a constant, i.e., the Bernoulli equation is “globalized.”¹⁸¹

$$m_j u_j^2/2 + 5kT_j/2 + q_j \phi = \text{const} - \Omega P_{j0}. \quad (21)$$

(The extra term on the right side appears if the global angular momentum is also constrained, Ω is its Lagrange multiplier.) An important consequence of Eq. (21) is that a finite- β relaxed state can only exist if there is flow, and in fact flow shear. The temperature (and hence the pressure) can only fall toward the plasma boundary if u_j rises toward the edge, i.e., sheared flow.

g. FRC-relevant solutions. The foregoing principles, if valid, apply in principle to any magnetically confined plasma. Several analyses have addressed the structure of supposed SO FRCs. These include solutions in one-dimensional “slab” geometry,¹⁸¹ two-dimensional uniform-density,⁶⁰ and solutions as the sum of two Chandrasekhar-Kendall functions (eigenfunctions of the curl operator).¹⁹⁶ One of the foregoing analyses introduced the concept of a helicity space $\{K_e, K_i\}$ to represent all SO-FRC states, with allowed and forbidden regions.⁶⁰

C. Global-tearing-stability principle

Suppose there is a vigorous instability with a well-defined marginal stability condition. Suppose further that frictional effects (e.g., resistivity) cause the plasma to evolve across the stability boundary. The instability then may act to quickly restore the profile at least to marginal stability. This is the essence of the global tearing-stability principle proposed quite some time ago,¹⁶⁰ which is related to the even older “delta-prime” method. It is applied as follows. Expand the magnetic flux function $\psi(r, z)$ as the sum of an equilibrium part and an axisymmetric perturbation $\psi_0(r, z) + \delta\psi(r, z)$, expand the equilibrium equation (viz. Grad-Shafranov equation) accordingly, and solve the perturbation-order equation $\delta\psi(r, z)$. If the solution has a zero *inside* the conducting wall (excluding, of course, the zero at $r = 0$) then the equilibrium $\psi_0(r, z)$ is tearing unstable; if the zero is at the wall then it is marginally stable; and if there is no zero inside the wall then it is tearing stable. Lately, this theory was applied to FRCs and styled as “tearing relaxation.”¹⁶¹ The results of this study are discussed later under the heading of experimental evidence for SO.

The conducting wall plays an essential role in tearing stability. If the wall is too far away then the first zero of $\delta\psi$ lies inside the wall—i.e. instability. If the wall is close enough, then stability is assured. The effect of the instability is to restructure the FRC by flattening the current density profile near the magnetic axis, thereby dissipating some of the poloidal flux $\phi_p (= 2\pi\psi_p)$. Thus, global tearing may regulate the decay of ϕ_p ; this possibility is discussed further in Sec. V.

The global-tearing-stability principle differs from the variational principles in that it is *essentially* instability driven. Even so it qualifies as SO because it maintains the plasma close to a “marginal” condition determined by a global condition. Tokamak sawtoothing is similar because it too is a tearing instability and it produces a non-disruptive restructuring. Sawtoothing is different because it is not regulated by a *global* stability criterion. Further, it occurs as discrete events well separated in time, whereas tearing relaxation in FRCs is likely to be a continuous “metabolic” process.

D. Dynamics of SO in FRCs

a. Transition. Like the Taylor principle, both SO principles discussed in the foregoing are “physics shy” in that they say nothing about the transitional dynamics, except that certain global quantities are roughly preserved. The question then remains—what “channel” might the FRC take in relaxing

toward a SO state? Simulation of the transition in an FRC is difficult because of the three-dimensional fine-scale structure and high frequencies associated with the expected turbulence. Of related interest are single-fluid resistive-MHD simulations in an idealized rectangular geometry.^{197,198} These observed selective decay of magnetic energy relative to magnetic helicity, leading to a state that is clearly not a force-free (“Taylor” state). Indeed, the perpendicular current ended up comparable to the parallel current. PIC simulations in an FRC geometry also showed a major restructuring without disrupting confinement.¹⁵⁰ The transition dissipated a substantial fraction of the magnetic flux.

Based on experience with RFPs and spheromaks, it is likely that low-order kink modes ($m = 1$, $n = 1$ or a small number) will be the primary engines of the transition and that it will be a mode-coupling process producing topological changes: surface breaking and rehealing. An ideal kink instability was the primary mechanism in the MHD simulations.^{197,198} The kinetic simulations,¹⁵⁰ which tracked the $n = 0 - 4$ structure, exhibited strong but transient multi-mode instability. Although neither simulation led to a force-free state, the MHD simulations led to a state with no persistent flows.^{197,198} Reports of the kinetic simulation made no mention of equilibrium flows.¹⁵⁰ Thus, the question remains open whether persistent flows can appear in a SO state.

b. Stability of SO states. An intuitive argument states that the chaotic transitional process should give free reign to every instability “in the book,” and thus, if not causing total disruption, lead to a stable state. Unfortunately, proofs of this are hard to come by, the lone exception to which is for strictly force-free states.^{170,199} Nevertheless, stability of more general SO states has been addressed occasionally, e.g., Ref. 200, considering second variations of the energy,¹⁷⁶ Lyapunov approaches,²⁰¹ and angular momentum conservation.²⁰²

c. Durability. If the SO state is stable, the question remains, will it retain its identity under the influence of frictional effects. The single study addressing this issue showed that in helicity space $\{K_e, K_i\}$, the helicities will evolve so as to preserve an FRC-like configuration.¹⁹⁶

E. Experimental evidence of SO in FRCs

Both RFP and spheromak plasmas show a clear natural tendency to relax to a Taylor-like state, at least in the plasma core. Evidence of SO is much less clear in FRCs, though several of its markers have appeared in experiments. None of these offer a convincing stand-alone proof for relaxation, but taken together are at least indicators thereof. Notably, the supposed SO states have very high β and are clearly not “Taylor” states. The evidences fall into two categories: markers of a transition from an initial to a relaxed state and appearance of preferred plasma structures.

a. Markers of a SO transition. There are several markers of a SO-transition, i.e., a dynamical phase producing a restructuring rather than the dissolution of the plasma: (1) *Survival of a violent birth:* The ejection of a plasmoid at

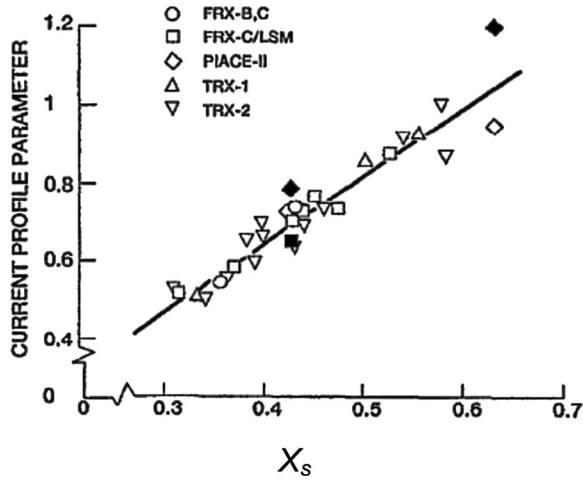


FIG. 12. Profile consistency in elongated FRCs. Reprinted with permission from L. C. Steinhauser and A. Ishida, *Phys. Fluids B* **4**, 645 (1992). Copyright 1992, American Institute of Physics.

super-Alfvenic speeds, its capture in a confinement chamber and its rapid settling down into a quiescent FRC has been observed in several “translation-trapping” and “collision-merging” experiments, as will be discussed in Sec. VI. (2) *Transitional dynamics*: Significant but transient low-order kink-mode activity ($n=1$ “tilt” as well as $n=2,3$) was detected by Mirnov probe arrays in elongated FRCs. These modes emerged early but subsequently died away as the plasma settled into a quiescent FRC equilibrium.^{124,125} (3) *Preservation of invariants*: Translation-trapping experiments producing an elongated FRC roughly preserved the magnetic helicity K_m .⁸⁹ On the other hand, in some “merging-spheromak” (Re Sec. VI) experiments, producing an oblate FRC, the initial K_m was dissipated in the transition.²⁰³ (4) *Flux conversion*: Translation-trapping and collision-merging experiments showed poloidal magnetic flux amplification factors of 1.5 (Ref. 89) and possibly much higher,¹⁷ while the toroidal flux decreased by comparable amounts. Somewhat different was a spontaneous toroidal-field-generating event during the equilibrium phase of a rotating-magnetic-field based FRC (Re Sec. VII).²⁰⁴

b. Preferred structures. FRC experiments exhibited a tendency to fall into certain natural or “preferred” states. (1) *Profile consistency* was observed in elongated FRCs, i.e., a consistent pattern in the current density structure.⁹⁷ Figure 12 shows a definite correlation between the current density structure and the radius ratio X_s . Here, the current profile index is $h \equiv (j_0/r)_0 / \langle j_0/r \rangle$, where “0” denotes the O-point value and the average $\langle \dots \rangle$ is over the midplane cross-section inside the separatrix. These results were found using transport modeling of the several experiments. Interpreting this pattern as evidence of a SO state has been questioned by the suggestion that it may simply be an equilibrium property of elongated FRCs in a flux conserver.^{50,51} (2) *Bifurcation of states*. Merging-spheromak experiments showed a clear bifurcation of final states depending on the initial magnetic helicity content.¹³⁰ If the K_m magnitudes of the two initial spheromaks were unequal and differed by less than 20%, the merged object transitioned into an FRC with $K_m=0$, other-

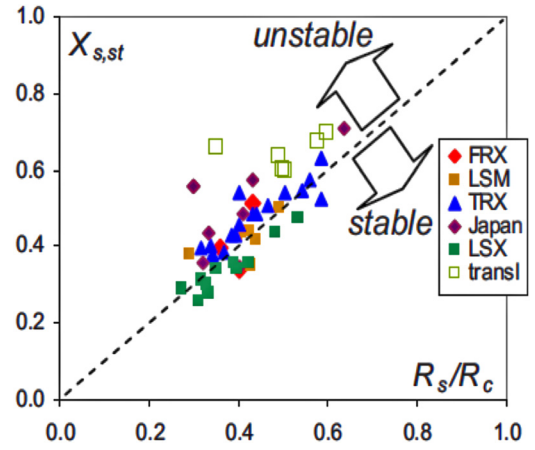


FIG. 13. (Color online) Global tearing stability: comparison of theory “ $X_{s,st}$ ” with experiment R_s/R_c . Reprinted with permission from L. Steinhauser and D. C. Barnes, *Phys. Plasmas* **16**, 092505 (2009). Copyright 2009, American Institute of Physics.

wise, the merged object became a spheromak. At larger- S_* , however, the range of conditions leading to an FRC became much narrower.^{28,29} (3) *Structure corresponding to a variational principle*. Two-fluid equilibrium modeling of translation-trapping experiments showed surface functions consistent of those for a minimum-energy state.⁶² This pattern applied in the core but not at the edge, just as is the case for RFPs and spheromaks where the “eigenvalue” $\lambda = j/B$ is uniform in the core but not at the edge. (4) *Tearing relaxation*. Perhaps, the strongest experimental evidence for SO regards the tearing-relaxation (global tearing stability principle).¹⁶⁰ Interpretation of experiment using the 2PE equilibrium paradigm (Sec. II) showed an agreement between the predicted tearing-stable radius $X_{s,st} \equiv (R_s/R_c)_{stable}$ and the experimentally measured value, as shown in Fig. 13;¹⁶¹ the predicted marginal-stability condition is the dashed line. This suggests that tearing stability regulates the equilibrium structure so as to maintain it close to the marginal-stable boundary.

V. TRANSPORT

A. Uniqueness of FRC transport

FRCs present unique challenges to the understanding of transport and confinement. In large part, this is the result of very high β (and the paucity of relevant transport theory), the large the ion Larmor radius, and the profound influence of the scrape-off layer (SOL) on global confinement. Even so, the understanding of FRC transport has advanced considerably since 1988. Note the following salient properties that have emerged, some of which are unique among magnetic confinement concepts. (1) Most elongated FRCs are quite isolated from material boundaries and have very low-recycling. (2) These FRCs have comparable particle, energy, and magnetic flux confinement times. (3) Much less is known about transport in oblate FRCs; experiments have focused on other issues. (4) Electron thermal loss is convective, rather than conductive, and radiative loss is relatively small in low-recycling FRCs. (5) Self-organization (SO) dominates in the core region so that the apparent transport is not the result of a local resistivity. (6) Cross-field particle transport in the

edge is a fraction of Bohm, at least superficially, although tentative evidence suggests that it has drift parameter scaling and may be gyro-Bohm or better. (7) The core temperature profile is quite flat.

Large questions about FRC confinement remain. (1) While empirical scaling laws have been extracted for *elongated* FRCs, they do not apply to all experiments, some of which perform somewhat better. (2) The cross-field transport mechanism in the pivotal edge region has not been determined from current experiments, much less how it extrapolates to hotter, fusion-grade conditions. (3) End-flow dynamics in the scrape-off layer and how it interplays with the cross-field transport remains unclear.

Unique as FRCs are, it is hardly surprising that a terminology has developed which differs to some degree from that used elsewhere. The *particle confinement time*, τ_N , in low-recycling FRCs is simply the decay time of the total particle inventory. The *flux loss time*, τ_ϕ , is the decay time of the poloidal flux ϕ_p ; most FRC experiments have no current drive so that this reflects the decay of the toroidal current. The *energy confinement time* τ_{Ep} is the decay time of the global plasma energy; it differs from the usual energy confinement time, which subtracts off “work” (from magnetic compression) and radiative losses. These effects are modest in low-recycling FRCs so that τ_{Ep} is a reasonable representation of the energy confinement.

The remainder of this section is organized as follows: (B) basic experimental observations of confinement and interpretive methods developed to extract transport rates therefrom; (C) proposed cross-field transport mechanisms in theory and experiment; (D) flow dynamics in the SOL; and (E) evidence for an “integrated” view of transport in FRCs.

B. Experiment and interpretation

a. Empirical scaling. A significant data base has been developed over the years for confinement in elongated FRCs, known as the “FRC data compendium.” This continues to be augmented and now includes recent results from high-density FRCs.^{27,84,206,208} Its most up-to-date form is available online.²⁰⁷ Empirical “laws” for FRC confinement date to the early 1980s with the venerable “ R_0^2/ρ_i ” particle confinement scaling

$$\tau_N = \text{const} R_0^2 / \rho_{i0} \quad (22)$$

reflecting results from FRX-B and C.⁷⁹ Since then various empirical scalings have appeared^{208–210} as well as the oft-quoted “LSX” scaling^{98,211} for the flux decay time

$$\tau_\phi = \text{const} X_s^{1/2} n^{0.53} R_s^{2.14}. \quad (23)$$

The appearance of X_s reflects an attempt to capture varying “profile” effects, since it is related to the amount of poloidal flux (Sec. II). Confinement of FRCs formed by translation methods (translation-trapping, collision-merging, Re Sec. VI) have often shown “better-than-LSX” scaling by as much as a factor of several.^{16,212} For example, Fig. 14 shows results from C-2 with flux confinement times up to a factor of 5 longer than LSX scaling.

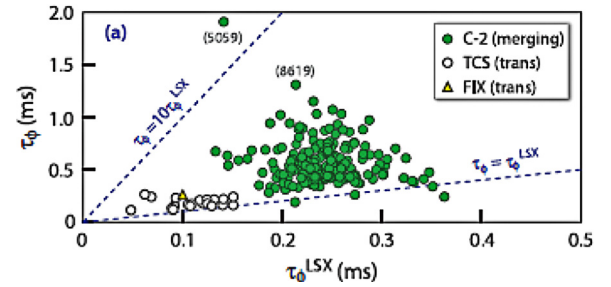


FIG. 14. (Color online) Comparison with LSX scaling for the C-2 experiments, Re Fig. 7 from Ref. 16. Reprinted with permission from M. W. Binderbauer, H. Y. Guo, M. Tuszewski *et al.*, Phys. Rev. Lett. **105**, 045003 (2010). Copyright 2010, The American Physical Society.

Attempts to get “closer to the physics” have occasionally prompted quotation of a gross transport rate, e.g., $\chi_N = R_s^2 / 8\tau_N$, but these neglect profile effects. As such, note that Eqs. (22) and (23) (the latter if $\tau_N \sim \tau_\phi$) are akin to the so-called “Alcator” scaling. The usual first step to draw closer to the physics has been to extract a resistivity from data. Given τ_N and τ_ϕ , global confinement models can be used to find the gross internal magnetic field structure and resistivity profile.^{53,82,213} By suitably adjusting the assumed resistivity profile any combination of τ_N and τ_ϕ can be reproduced, allowing one to infer the resistivity and its gross profile from experimental data.

b. Longitudinal effects. A missing element in transport analysis has been a “longitudinal” study, to borrow a medical term, i.e., how the transport rates evolve during the lifetime of a decaying FRC. In this regard, it has sometimes been noted that the separatrix radius exhibits a linear rather than exponential decay in time.^{104,105} This suggests a transport rate evolving with the changing plasma conditions (R_s , X_s , etc.). Relevant to this, simulations of FRC formation often use the *Chodura resistivity* (empirically-adjustable resistivity²¹⁴). Some of these computations were carried into the equilibrium phase; the same Chodura coefficients that worked well in the formation phase gave a reasonable representation of the subsequent confinement. It is interesting to compare Chodura to the Bohm transport rate. A modified Bohm rate D_B^* is

$$D_B^* = kT_i / 32eB_e. \quad (24)$$

This differs from the traditional Bohm rate by using T_i instead of T_e and B_e instead of B ; they are identical if $T_i = T_e$ and $B \sim B_e$. For a drift parameter $V^*/c_s < 1$, the Chodura rate is $\chi_{Ch} \propto D_B^*(c/c_s)V^*/c_s$ where $V^* \equiv j/en$ is the relative ion-electron drift, c is the speed of light, and $c_s \propto T_i^{1/2}$ is the speed of sound). Thus “Chodura” is similar to “gyro-Bohm” except for an extra factor $\propto T_i^{-1/2}$.²¹⁵

c. Equilibrium-based interpretation. An interpretive method that is easier to apply exploits unique properties of elongated FRCs. Using the “two-point equilibrium (2PE) paradigm (Re Sec. II) enables extraction of transport rates in the FRC data compendium,⁸⁴ allowing one to get closer to the physics by inferring the transport rate in the critical separatrix region. Figure 15 compares the inferred particle

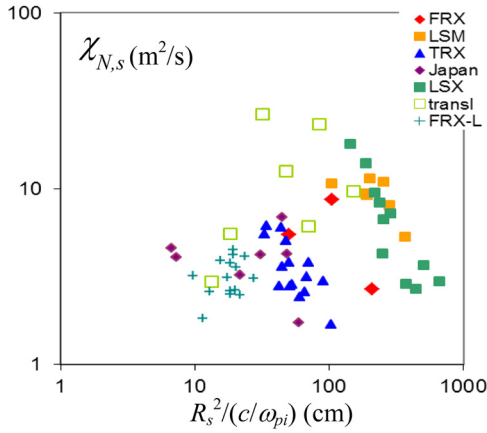


FIG. 15. (Color online) Edge particle transport rate for elongated FRCs as inferred by the 2PE interpreter. Here “LSM” represents the FRX-C/LSM experiments and “transl” translation-trapping experiments.

transport rate at the separatrix (Re Eq. (1) of Ref. 216) to a variation of the traditional R_0^2/ρ_{i0} scaling, Eq. (22), which here is modified to $R_s^2/(c/\omega_{pi})$. Since the ion skin depth $c/\omega_{pi} \approx 2\rho_{i0}$, the two reference scalings are about the same. In Fig. 15, observe that the edge transport rate is typically in the 2–20 m²/s range. Yet, there is significant scatter: factors of five within individual facilities and factors of ten for the full data base. Clearly, the scalings like Eq. (22) are an unreliable predictor of how $\chi_{N,s}$ varies.

If the $\chi_{N,s}$ were plotted against the Bohm rate D_B^* , most of data would lie in the range $\chi_{N,s}/D_B^* \sim 1/10 - 1/3$. Thus, the transport rate could be Bohm-like or gyro-Bohm, as hinted earlier. This question will be revisited in Sec. V C.

d. Other models. Zero-D transport models occasionally appear, especially for representing the power balance (e.g., in high-density FRX-L).²⁰⁶

C. Cross-field particle transport

The effort to identify the transport mechanism in FRCs has a long history. The focus has been drift turbulence driven by free energy associated with the drift parameter V^*/c_s . Inferred drift parameters at the separatrix are $V^*/c_s \sim 0.6 - 0.8$ (Re Fig. 4 of Ref. 84). The region of this moderate level of drift is fairly narrow; V^*/c_s falls rapidly toward the plasma core, typically by a factor of five or more for $r = 0.9R_s$. Thus, traditional drift-turbulence mechanisms may not be active in the core region. Several questions then emerge in the discussion of transport: (1) what is the mechanism in the narrow, high-drift edge region; (2) what is the effect of high $\beta \sim 1$; (3) what is the mechanism in the low-drift core; (4) does a global principle of some sort regulate the core transport; and (5) how is cross-field transport in the core and edge affected by rapid parallel outflow in the SOL?

a. Background—classic LHD. The earliest candidate for transport mechanism is the lower-hybrid drift (LHD) instability. Originally developed to explain cross-field transport in θ -pinches,²¹⁷ it became a popular possibility for reconnection in space plasmas. The traditional LHD instability is

electrostatic ($\delta B = 0$), flute like ($k_{||} = 0$), with wave numbers $k \sim 1/\rho_e$ (ρ_e = electron Larmor radius), and has a transport rate of the form (hereafter called “drift scaling”)²¹⁸

$$\chi_N \propto D_B^* (V^*/c_s)^2, \quad (25)$$

i.e., Bohm-like but proportional to the drift parameter squared. However, classical LHD transport has not fared well in FRCs. It appears not to explain the overall transport rate.^{209,213} Finite- β stabilization of LHD (Refs. 219,220) means that it cannot explain the anomalous loss of magnetic flux,²²¹ which, if interpreted as a resistive effect, occurs at the O-point. Moreover, attempts to detect the fluctuations have found that the observed $\delta n/n$ values are two-orders of magnitude lower than predicted by LHD theory.^{82,120,222} Moreover, the observed fluctuations are electromagnetic, not electrostatic.^{120,220} A partial exception to this are observations in a reconnection layer (MRX): here electrostatic fluctuations were observed, but only in the low- β periphery of the current sheet, and they appeared not to explain the reconnection rate.²²³ The obvious question is why does LHD turbulence not appear. Two possibilities have been investigated. One is that LHD fluctuations couple strongly to lower- k (wave number) modes, which serves to damp the LHD.²²⁴ The other is that an electromagnetic version ($k_{||} \neq 0$, $\delta B \neq 0$) might be active,²²⁵ a possibility that has resurfaced lately.²²⁶ In what follows, recent measurements of fluctuations are reviewed, followed by theories addressing various drift-turbulence mechanisms.

It should be noted that an early treatment of drift turbulence in a collisionless, high- β plasma ($\nabla p = 0$) predicted transport rates up to ten times Bohm.²²⁷ Such high rates certainly have not been observed in FRCs, where $\chi/D_B^* \sim 1/10 - 1/3$.

b. MRX Experiment Fluctuations near the center of a magnetic reversal layer in MRX (Ref. 228) were electromagnetic, obliquely propagating ($30^\circ - 60^\circ$ angle between $\mathbf{k}$ and $\mathbf{B}$) with phase velocity comparable to V^* and with $k_{\perp}\rho_{i0} < 20$. The fluctuation amplitudes correlated well with the enhancement in reconnection rate (compared to classical resistivity). The fluctuations resembled that expected for the modified two-stream instability, which is unstable at high- β in contrast to the traditional LHD. The drift parameter (referenced to the Alfvén speed) was $V^*/V_A \sim 5$ (corresponding to $V^*/c_s \sim 4$); this is well above inferred maximum values in FRC experiments ($V^*/c_s \sim 0.6 - 0.8$).

c. Drift turbulence—kinetic modelling. A kinetic analysis²²⁹ focused on low-frequency ($\omega < \omega_{ci}$) fluctuations since they are more likely to be global, producing large-scale motions and, consequently, faster transport. This was an attempt to understand the anomalous flux loss associated with the low- V^*/c_s core region. It addressed the high- β analogy of drift-dissipative modes, i.e., electromagnetic turbulence destabilized by finite collisionality or trapped-particle modes. An estimate of the resulting resistivity roughly agreed with experimental values of τ_ϕ .²³⁰ These results, however, are controversial; a later study indicated that these “low-frequency drift”

waves should not, in fact, be unstable.²³¹ A recent kinetic treatment considered electromagnetic drift instabilities assuming $k_{\parallel} \ll k_{\perp}$, but allowing nonadiabatic responses of both ions and electrons.²³² Instability driven by gradients of density and temperature and well as field-line curvature were addressed. Unstable growth was observed for large ∇T_e ($\eta_e \equiv |\nabla \ln T_e|/|\nabla \ln n| \sim 1.5 - 2$) and ∇T_i ($\eta_i > 1$). The mode appeared to be stable for $\eta_i < 1$. In view of the large ρ_i in FRCs, it is unlikely that η_i exceeds unity. Moreover, limited experimental evidence suggests that $\eta_e \sim 0.25$ in the core and perhaps as high as unity in the SOL.⁸⁴

d. Drift turbulence–collisional plasmas. A fluid treatment of drift instabilities addressed high-densities relevant to magnetized-target fusion (MTF) as investigated in FRX-L.²³³ This followed conventional assumptions in drift wave theory²²⁷ albeit with modifications for high collisionality. The predicted transport rate was $\sim D_B/2$ for MTF-relevant parameters.

e. Drift turbulence–two-fluids. A two-fluid theory considered conditions relevant to the current sheet center in MRX ($V^*/c_s = 5$), considering intermediate frequencies $\omega_{ci} < \omega < \omega_{ce}$, and unmagnetized ions ($k\rho_i > 1$).²²⁰ The waves were destabilized by a reactive coupling between the backward-propagating whistler waves (electron frame) and the forward propagating sound waves. They propagate obliquely $k_{\parallel} \sim k_{\perp}$, with $k_{\perp}\rho_i \sim 5 - 15$. Large V^*/c_s produced a transport rate $\chi > 1000 \text{ m}^2/\text{s}$ much higher than the $2 - 20 \text{ m}^2/\text{s}$ levels in FRCs (Fig. 15).²³⁴ A recent treatment using the same model²²⁶ recognized that the unstable modes had significant “radial” group velocity, causing them to propagate out of the current sheet before significant growth could occur. However, another electromagnetic mode with $k_{\parallel} = 0$ and a small group velocity were discovered. This analysis was also for large $V^*/c_s \sim 2.5$.

f. Drift turbulence–simulation. All the foregoing were local linear theories. Two-fluid simulations have investigated global nonlinear behavior in current-sheet and FRC-like geometries.²³⁵ In an edge current sheet of an infinitely elongated FRC (analogous to the SOL), the nonlinear growth of fluctuations appeared with $k_{\perp}\rho_i \sim 6$, this despite beginning the simulation with a much lower $k_{\perp}$ initial disturbance. Another set of computations addressed a Harris sheet with half width $\sim c/\omega_{pi}$, corresponds to a uniform $V^*/c_s \sim 0.8$. Instability appeared initially toward the edge of the sheet where the density gradient was maximum. Eventually, though, the fluctuations penetrated inward to envelope the current sheet center. This arrangement differs from FRCs where the drift parameter is highly nonuniform. Also, the artificially low mass ratio $m_i/m_e = 25$ means that $\rho_e = \rho_i/5$, making it hard to distinguish between low frequency modes with $k_{\perp}\rho_i \sim 1$ and LHD-like modes with $k_{\perp}\rho_e \sim 1$.

g. Other effects. (1) Sheared $E \times B$ flows play an important role in reducing turbulence in high-temperature plasmas.^{236,237} Further, a naturally large flow shear occurs near the FRC separatrix. This can be inferred from the anticipated current density profile (Fig. 3) together with the recognition

that the ions probably carry most of the current. (2) Translation-trapping formation methods (Sec. VI) naturally produce a modest toroidal field. These FRCs show better confinement by some measures.¹⁰³ The toroidal flux decay time appears to be comparable to that for the poloidal flux.²² (3) The presence of a “neutral bath” in the confinement section increases the apparent particle confinement time.²³⁸ (4) Low-density FRCs formed by rotating magnetic fields exhibit two significant differences from θ -pinch-based FRCs: relatively large transport rates ($100 - 200 \text{ m}^2/\text{s}$) (Re. Fig. 15) (Ref. 239) and large recycling and impurity pickup. Measures to improve plasma hygiene produced markedly lower impurity radiation loss and improved confinement.^{240,241}

h. Summary of particle transport. A variety of theories have been advanced to explain the particle loss and flux decay in FRCs. These suffer from various assumptions that are not relevant to FRCs, whether too-high drift parameter, or too steep temperature gradient. The effects of flow shear on drift wave stability have not been investigated in the FRC context.

D. Thermal transport

The SOL is exposed to rapid thermal loss along field lines to external walls. However, there is also the possibility that *all* magnetic surfaces, including those in the core, may in fact be opened. This definitely is the case for rotating-magnetic fields (RMFs) a current drive technique employing an oscillating transverse field (Sec. VII). The same occurs if multipoles are applied to stabilize the rotational instability (Sec. III). Finally, natural turbulence is likely to “break” magnetic surfaces (Sec. II). Indeed even infinitesimal $\delta B_{\perp}$ perturbations can lead to large radial excursions of the field lines in FRCs (“Field-line wandering”) with $\delta r \sim R_s$.²⁴² Electrons, highly mobile along field lines, are quick to respond to any such “opening.” Any of these effects converts the nominally closed magnetic structure into one which is open to electron thermal loss along field lines.

Despite this foreboding possibility, experiments suggest that electron thermal loss along open field lines is limited to convective rates, much slower than would be the case for parallel thermal conduction. Results from both θ -pinches and FRCs show that the open-field-line end-loss of energy is convective for *all* ranges of collisionality, i.e., parallel electron thermal conduction is never the controlling process.²⁴³ The ambipolar potential serves to maintain electron confinement. This corresponds to the so-called sheath-limited regime.²⁴⁴ As a result, the electron energy loss per electron-ion pair escaping system is $W_e \sim f k T_e$, where typically $f \sim 4 - 8$, much lower than would be the case for thermal conduction loss. The reason is that thermal conductivity simply does not apply if the connection length is less than about $100\lambda_{ee}$ (mean free path). Core thermal loss should also be convective. If anything slightly lower values of $f = W_e/kT_e$ may occur because of longer connection length. Convective thermal loss is the likely explanation why the energy confinement time τ_{Ep} is always about $1/2\tau_N$ in elongated FRCs.

Critical to the convective thermal loss property is the relative isolation of an FRC from external boundaries and

low recycling. Here, considerations that apply to thermal loss mirror plasmas have been explored at length.²⁴⁵ A key similarity between FRCs and mirrors is that there is no restriction of the size of large expansion chambers at the ends, i.e., the so-called “natural divertor.”

Convective thermal loss has several implications. (1) The FRC is genuinely an isolated system, i.e., not subject to thermal-conduction loss to cold boundaries, which, by the way, would be extreme at high temperature. (2) The burden of energy confinement is shifted to the question of particle confinement; improved τ_N should lead to improved τ_{Ep} . This holds as long as recycling can be kept small. (3) Self-organization (SO) processes in the core may not compromise energy confinement. A well-known downside of SO is the breaking of magnetic surfaces and enhanced energy transport. This, however, is not an issue if $\tau_{Ep} \sim \tau_N$. (4) The SOL can run “hot” because of the natural divertor and the unlimited divertor plate area.

E. Scrape-off layer (SOL)

The SOL must be treated as an important element in the overall transport picture. This is not merely a “cold boundary” condition. Indeed it is not: temperature profiles in experiments indicate that the SOL, at least at the separatrix is nearly as hot as the core plasma. Further, unlike in the core where transport is a cross-field process, in the SOL it is a balance between cross-field diffusion and parallel flow. An early global transport analysis recognized this coupling.²⁴⁶ This model incorporated two assumptions that might be questioned: (1) that the cross-field mechanism is known (LHD in this case) and (2) that the outflow rate is known (sonic speed in this case). The first assumption was subsequently challenged as reviewed in Sec. V C. A subsequent analysis of experiments challenged the latter, as described in the following:

a. Observations. While FRC diagnostics mainly give gross properties, a few have yielded information about detailed features such as the SOL thickness. Some of these were analyzed 25 years ago, leading to the conclusion that the SOL is either anomalously thick or else the end-flow rate was anomalously slow, i.e., a factor of several lower than the speed of sound.¹¹⁹ A recent review of old and new SOL data confirmed the anomalously broad SOL (Re Table III of Ref. 85). These observations fostered the proposal that a natural electrostatic potential barrier anchored near the X-points significantly reduces the escape of ions, Re Fig. 3 of Ref. 119.

Other intriguing observations have also appeared. (1) The magnetic flux “entrained” in the SOL was somewhat larger than that in the jet, i.e., a “jet narrowing” effect; this indicates that the outflow of ions did not simply follow the magnetic field lines.¹¹⁹ (2) Based on limited results from FRX-C,²⁴⁷ a transitional structure appeared in the jet (divertor) region suggestive of a double layer separating the hot, outflowing plasma from a cooler plasma behind the layer.¹¹⁹ (3) Electrostatic probe measurement in FIX (Ref. 158) prompted the perplexing suggestion that the end flow is not anomalous at all, but in agreement with sonic flow. However,

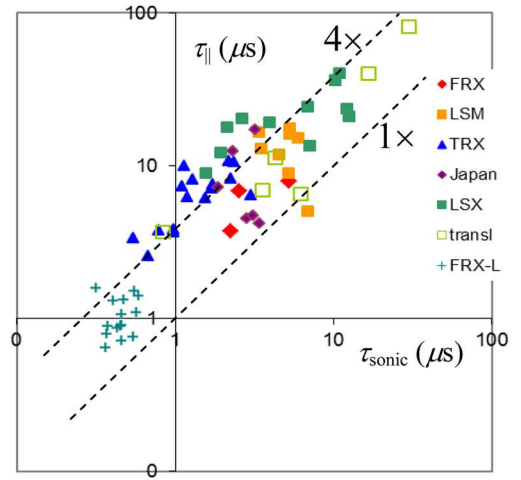


FIG. 16. (Color online) SOL end-loss time compared to sonic time.

the FRC data compendium, Re Fig. 16, suggests that the inferred SOL end-loss time $\tau_{||}$ (Eq. (2) of Ref. 216) is somewhat longer than the sonic time $\tau_{sonic} \equiv L_s/2c_s$ and by factors of 4 – 6 in some cases. Even so, there is considerable variation from one experiment to another. Note that FIX has strong mirrors at the ends that likely have a strong influence on the outflow.¹⁵⁸ Several models of SOL end flow have emerged in an attempt to explain these anomalies.

b. VSPL models. Perhaps the first such model was the velocity-space-particle-loss (VSPL) mechanism, an analogy to the loss-cone confinement criterion in mirrors.²⁴⁸ This kinetic approach led to a simple confinement criterion⁷³ in terms of the constants of motion of individual ions

$$H + \omega_{cj} P_\theta < 0 \quad (26)$$

The two constants of motion are the Hamiltonian, $H = mv^2/2 + q\phi$, and the canonical angular momentum, $P_\theta = q\psi + mrv_\theta$, and $\omega_{cj} = eB_j/m$ is the ion cyclotron frequency at the maximum field point ($B = B_j$) in the jet. Equation (26) includes the electrostatic potential effect (through H), but early applications of it to the end-loss problem assumed $\phi = \text{const}$, e.g., Ref. 73. After the recognition of anomalously slow end loss, extensions of the VSPL model have included electrical potential effects.⁷⁴ A one-dimensional model assumed a magnetic field structure and distribution function corresponding to a rigid-rotor (Sec. II) and found the self-consistent ϕ profiles required by charge neutrality.⁷⁵ The model also allowed finite confinement time of unconfined ions (non zero $\tau_{||}$). As such it recognized the interplay between the collision time τ_c by which ions “random-walk” across magnetic surfaces and their end-loss time $\tau_{||}$ once they are unconfined. The same approach was later extended to two-dimensions in order to ascertain whether an end-loss retarding potential structure appeared.⁷⁶ The results showed a significant potential peak just downstream from the X-points. Further, the jet-narrowing tendency appeared, largely a result of the fact that a significant fraction of ions in the SOL are, in fact, confined according to Eq. (26) and as such are not immediately subject to end loss.

c. Fluid models. The traditional divertor-plasma description is the “magnetic duct” model, widely used in tokamaks.²⁴⁴ This is readily adaptable to an FRC although with the major addition of varying duct area. Then, the flow rate is regulated by a Laval nozzle effect, i.e., a sonic condition at the narrowest point in the duct, i.e., the maximum field point in the jet region. This approach has been applied to FRCs, including anisotropic ion effects²⁴⁹ as well as the effect of a strong applied mirror in the jet region.¹⁵⁸ However, the magnetic duct analogy is imperfect since “fluid” ions do not, in fact, flow along magnetic surfaces ($\psi = \text{const}$) but rather along their own surface functions $P_{i\theta} = \text{const}$ (Sec. II).⁵⁹ Indeed, the so-called separatrix, separating confined from unconfined regions is complex since the ion-fluid separatrix ($P_{i\theta} = 0$) lies *outside* the electron fluid separatrix ($\psi = 0$), Re Fig. 1 of Ref. 59. This will produce important electrostatic effects that may also influence the axis region of the FRC.

d. Kinetic models revisited. The possibility of a “complex separatrix” points to a fundamental weakness of fluid-based models. Indeed, even Braginskii’s gyroviscous correction,⁵² a fluid-based representation of finite-Larmor radius, is doubtful since it assumes $\rho_i \ll L_n$, which is hardly valid in the SOL. Thus, though fluid-based models are attractive, interest in kinetic effects continues, e.g., in the effect of collisionless (nonadiabatic) ion scattering near X-point.²⁵⁰ Moreover, the old idea of a realistic ion distribution recognizing rapid end loss of unconfined ions, originally developed in the context of an Astron,⁶⁴ has re-emerged as relevant to FRCs.⁶⁹

F. Integrated transport

a. Hybrid confinement. The coupling of interior-diffusion and SOL-flow processes was recognized earlier²⁴⁶ and led to a “hybrid” confinement time depending on both χ_n and $\tau_{||}$. Moreover, the action of non-resistivity-driven processes dependent on global considerations, e.g., “tearing relaxation”¹⁶¹ also enters an “integrated” view of transport. The emerging picture of FRC confinement includes the following: (1) Energy confinement (τ_{Ep}) is convective and “slaves” off the particle confinement (Sec. V D). (2) Flux confinement (τ_ϕ) is determined by a global “profile consistency” principle rather than the resistivity at the O-point; consequently, it too “slaves” off the particle confinement (Sec. IV). (3) Particle confinement (τ_N) is the interplay between the cross-field transport rate at the separatrix ($\chi_{n,s}$) and the end-loss rate ($1/\tau_{||}$) in the SOL (Eq. (2) of Ref. 216). (4) Since $\chi_{n,s}$ likely is driven by a local drift-turbulence, it depends to some degree on the local drift parameter at the separatrix (e.g., Eq. (25)), further coupling it to the profile steepening effect of $\tau_{||}$.

The coupling, of $\perp$ and $||$ processes was investigated recently,²¹⁶ assuming drift scaling (Eq. 25) but with an adjustable but universal (all data) constant of proportionality. The results are shown in Fig. 17. The scatter in the data was lower than a similar procedure comparing to Bohm or gyro-Bohm scaling. This supports the proposal that the transport mechanism is some version of drift-turbulence with a drift-scaling rate. If so, the particle confinement time is given by Eq. (2) of Ref. 216:

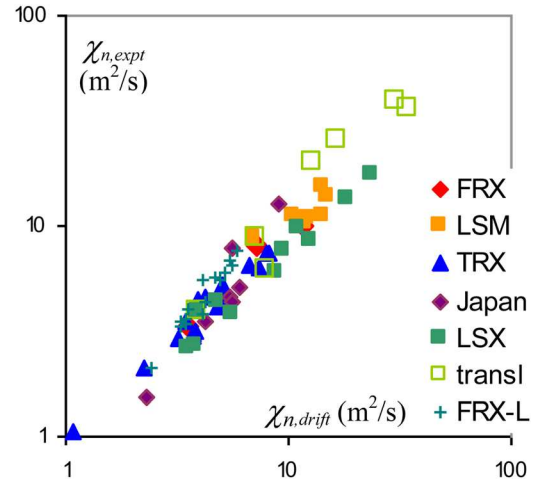


FIG. 17. (Color online) Comparison of edge particle transport rate to drift scaling.

$$\tau_N \approx 2(\langle n \rangle / n_s) S_* \left(\tau_{||}^3 / \omega_{ci,s} \right)^{1/4}. \quad (27)$$

This expression has profound and yet perplexing implications. It implies that the confinement is heavily dependent on parallel confinement in the SOL, in fact $\tau_N \sim \tau_{||}^{3/4}$, i.e., it is nearly linear with $\tau_{||}$. In effect, the “tail” (SOL end-flow physics) wags the “dog” (global confinement τ_N). If this trend holds, then the highest-leverage way to improve confinement is to reduce the end-loss rate in the SOL. This prompted the proposal of an “end-plugged” FRC (Ref. 216) using strong magnetic mirrors in the jet region in the spirit of the Gas Dynamic Trap.²⁵¹

b. Effect of controls on transport. Various controls have been proposed for FRCs to guarantee stability or sustain the plasma. These may affect the global particle confinement. FRCs rotate, whether as a result of beam injection (for heating or current drive)^{15,252} or naturally by electrical shorting effects (Sec. II). The rotation eventually drives a rotational instability, but it can be suppressed by applying multipoles fields (Sec. III). These, however, introduce a transverse field component $B_\perp$ that may compromise confinement. Indeed, strong multipoles do enhance the particle transport rate.²⁵³ A rotating magnetic fields, a proposed current-drive technique (Sec. VII), are essentially a rotating multipole and also introduces a $B_\perp$, although this might be mitigated by employing “odd-parity” field structure.²⁵⁴ The key question is how important this parasitic effect is in hotter larger-size FRCs. The scaling may be favorable since larger sizes have lower average rotation rates. On the other hand, introducing even a weak energetic beam component can produce improved confinement, as shown in FIX.^{166,255} These experiments used highly oblique beam injection, i.e., at a small angle to the geometric axis. Another control technique with the potential to reduce the flow shear in the edge and SOL region; this technique is at the heart of the Maryland Centrifugal eXperiment (MCX).²⁵⁶ The consequent velocity shear suppresses flute-like instabilities²⁵⁷ but may also suppress transport-driving turbulence.

VI. FORMATION

A. Introduction

The birthing of a magnetic confinement configuration is nearly as complicated and precarious as it is for biological species. It involves a complex recipe of steps, each with its own mysteries: gas injection, pre-ionization, and sequenced ramping of magnetic coils, all this in addition to continuous vacuum pumping and wall-preparation procedures. FRCs are no exception, but here the similarities end. Start-up of an FRC is called “formation” because it is assembled, as it were, layer by layer. Formation must generate a large plasma current comparable to and opposed to the current in the surrounding coil set. In most cases, this is accomplished without the benefit of a central “ohmic” coil. Moreover, the birthing process is extremely violent, usually involving Alfvénic speeds.

The traditional FRC formation method is by θ -pinch. One of the most striking areas of progress since 1988 (Ref. 5) has been the emergence of a variety of alternative formation methods as well as significant improvements on the θ -pinch method itself. Most of these technologies have been quite successful. What follows is a brief review of advances in the (1) θ -pinch method and then a description of the several innovative formation methods; (2) coaxial source; (3) translation-trapping; (4) collision merging; (5) spheromak merging, (6) rotating magnetic fields, and (7) energetic particle injection. Each has its own peculiar technologies, some quite intricate. The emphasis here is on the physics, with technological issues mentioned only in passing.

Traditional FRC formation uses a “ θ -pinch,” a linear, elongated magnetic coil surrounding the plasma chamber. The sequence, illustrated in Figure 18, is as follows. Discharging the “bias” capacitor bank creates a “reverse” magnetic field in the plasma chamber. One or more preionization (PI) methods generate and imbed a plasma in the bias magnetic field. Discharge of the “main” capacitor bank rapidly reverses the magnetic field. A current sheet, initially thin, shields the plasma from the rising “forward” field. Thus, the rising field acts as a magnetic piston to radially implode the plasma. Although the current sheet broadens, the plasma current remains large, comparable to the coil current. Upon reaching maximum current, the coil-capacitor system (LC circuit) is

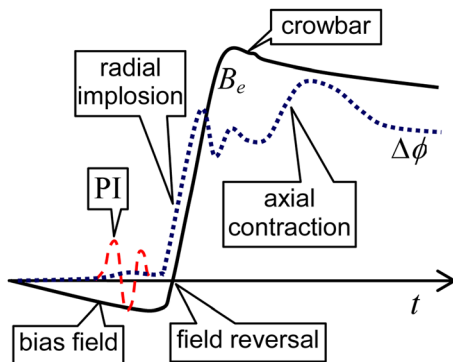


FIG. 18. (Color online) Sequence in θ -pinch formation of an FRC, showing the history of the PI current, external magnetic field B_e , and excluded flux $\Delta\phi$.

“crowbarred,” isolating the capacitor bank, and leading to a simple resistive decay of the coil current (LR circuit). In the meantime, the highly elongated plasma “reconnects” at each end to form closed magnetic surfaces with reversed field “inboard” of the magnetic null sheet, and forward field “outboard.” Strong magnetic curvature forces at the ends then drive a piston-like axial contraction. The radial and axial dynamics rapidly die away, leaving behind a quiescent FRC.

The salient features of the θ -pinch method are as follows: (1) *purely inductive*, an electrodeless formation method; (2) *large inductive voltages* (10’s of kV); (3) *fast formation*, with timescales of 2 – 20 μ s; (4) *highly dynamical*, with radial and axial flows comparable to the Alfvén speed; (5) *highly elongated* FRCs appear naturally, with elongations of 4 – 20; and (6) *highly isolated* FRCs also appear naturally, with plasma to-wall separation comparable to the separatrix radius.

B. θ -pinch improvements

a. Voltage reduction. High voltage, the major technological challenge of θ -pinch formation, is required to rapidly reverse the magnetic field. This is needed to shorten the interval when the plasma leans against the wall and can lose a large fraction of the reversed flux. A study of the wall current sheath²⁵⁸ showed that the flux loss is resistive diffusion rather than an Alfvénic flow to the wall, as predicted earlier.²⁵⁹ This insight motivated the TRX-2 experiments, which used a series of three-turn coils instead of the usual single-turn coil.²⁶⁰ The electric field E_θ associated with the rising field was lower by a factor of 3–6; less than 100 V/cm. Importantly, the plasma temperature did not suffer, although the slow field rise nearly eliminated radial shock heating, other effects compensated: ohmic heating from flux dissipation and axial contraction. Moreover, a slower field rise gave “room” for the application of other formation controls using auxiliary coils.

b. Programmed formation. Empirical scalings indicate that improved confinement depends on achieving a higher poloidal flux ϕ_p , Re Sec. V. The naive approach to higher- ϕ_p is to increase the initial bias field B_0 or the filling pressure p_0 . Unfortunately, both measures *increase* the violence of the axial contraction. Higher B_0 increases the field-line tension that drives the contraction. Higher p_0 reduces the temperature, weakening the viscosity, which is the primary factor mitigating the violence of the contraction. Various methods for this, collectively called “programmed formation” were studied in early experiments in the Soviet Union (Re Ref. 5), and more recently there,²⁶¹ in TRX-1 (Re Ref. 5), TRX-2,^{260,262} and LSX.²¹¹ The most notable such technique is the independent activation of short “trigger” coils at each end of the main coil.^{260,261} Delayed trigger-coil firing leads to a more gentle axial contraction.

c. Other methods. Residual non-uniformities left over from the PI plasma or fluting during the radial compression can cascade into large-scale (low- n) instabilities that either destroy the FRC or greatly reduce the equilibrium poloidal

flux. Various PI methods, usually in combination, have been applied to make the PI plasma more uniform. The most extensive tests were on LSX:²¹¹ glow discharge, microwave horns at each end, half-wave antennas at the ends, end cusps, pulsed barrier fields, ringing θ -current discharge, and z -current discharge. High-order multipole windings ($n = 12$) create a *barrier field*, a localized transverse magnetic field near the wall.²⁶² This acts as a barrier to reduce plasma-wall contact during field-reversal. In addition to improving the PI the symmetry, *barrier fields* also reduce impurity pick-up from the wall and improve the retention of reversed flux.

d. Increased poloidal flux. Larger FRCs suffer more severely from violent formation dynamics because it is harder to achieve PI uniformity and larger size weakens the viscosity effect. LSX addressed issues by applied elaborate formation recipes based on the techniques just described,²¹¹ achieving good confinement (the so-called “LSX” scaling²⁶³) for ϕ_p up to about 12 mWb.

e. Extreme-densities. A sphere of the FRC research emerging in recent years is magnetized target fusion (MTF). Here, the FRC becomes the target for a very strong compression, hopefully leading to fusion conditions with an initial plasma of modest-size. MTF favors high density FRCs. The FRX-L program²⁶⁴ has addressed the production of the high-density FRCs, achieving densities $n_e \sim 1\text{--}4 \times 10^{22}\text{m}^{-3}$ (about 10 times higher than past θ -pinches), and total temperatures $T_i \sim 200\text{--}400\text{ eV}$ (Ref. 84) with good shot-to-shot reproducibility.²⁰⁵ On the other end of the density scale, θ -pinch formation is limited at low-density by difficulty achieving adequate preionization: modifications to the PI method were to achieve lower-density operation were addressed on NUCTE.²⁶⁵

f. Modelling. Although θ -pinch formation involves multiple steps, simple analytic models have captured two effects: heating intrinsic to formation^{206,266,267} and flux retention.^{258,266} However, understanding details of the formation require at least a two-dimensional (2D) MHD simulation. The MOQUI code has served for many years as the work-horse tool for modeling FRC formation.²⁶⁸ No 2D simulation can replicate non-axisymmetric aspects of formation, such as the “flux dump” phenomenon²⁶⁰ occurring between the peak of the axial contraction and the equilibrium phase. Its effect can be replicated in 2D only by the artifice of an *ad hoc* high resistivity.²¹⁴ Another practical difficulty is that 2D simulations sometimes predict features for which no confirmatory diagnostics are available, e.g., an internal doublet structure.¹⁰⁰ A recent development is the LamyRidge code, a powerful new 2D-MHD code for modeling FRC formation; it includes a neutral gas, realistic boundary and initial conditions, and external electrical circuits.¹⁷

All the foregoing models are fluid-based. A kinetic simulation designed for stability studies has also modeled formation albeit with some artificialities.¹⁰¹ It exhibited one unusual feature, also not testable by current diagnostics: a persistent “up-down” internal toroidal field (TF) structure: positive TF at one end and negative TF at the other.

C. Coaxial source

Traditional θ -pinch methods require the forward magnetic field to rise faster than the resistive decay of the plasma current. While this decay is most rapid during field reversal (Re Fig. 18) when the current sheet rests briefly on the wall, it continues during the radial implosion as the current sheet moves inward. Inserting a *central solenoid* offers a means to mitigate the resistive-decay rate and reduce the required voltages. This flux core, or “ohmic coil,” delivers reversed magnetic flux to the inboard side of the plasma simultaneous with forward flux ramping up on the outboard side. Figure 19 illustrates this so-called “coaxial source” method. In principle, the Ohmic coil can inject a much larger amount of flux into the plasma than a θ -pinch and at a lower voltage. Finally, if the flux core is single-ended, as in the figure, the annular plasma can be ejected from the end of the flux core, delivering a conventional FRC. The Coaxial Slow Source (CSS) facility investigated this technique.^{269–272} Recently, the Irvine Field-Reversed Configuration (IFRC) facility,³⁷ aimed at development of flow diagnostics, employs it as well.

With relatively limited investigations, coaxial sources achieved a respectable poloidal flux $\phi_p \sim 5\text{ mWb}$. In terms of temperature, though, the non-dynamic formation proves a liability. The relatively low power to the plasma has resulted in temperatures of $T_i \sim T_e \sim 10\text{ eV}$, perhaps limited by radiation barriers. Low temperature may also result because the plasma is in more-or-less direct thermal contact with the wall, especially the inner wall, and may lack a separatrix.

D. Translation-trapping

a. Introduction. Two other formation methods use θ -pinches but with a significant difference: the pinch only acts as a source to create a high- β plasma before launching it into a separate confinement chamber. The first such method is *translation-trapping*. The pinch “source” is operated in the usual “reverse-bias” manner to imbed a reverse field in the plasma. The source is end-to-end asymmetric so that axial magnetic forces eject the plasmoid out one end. This motion, commonly called *translation*, is allowed because of two natural attributes of θ -pinch FRCs: no center column and a thick layer of open magnetic flux enveloping the plasma. These features allow the plasmoid to slide unimpeded along its geometric axis. Maintaining confinement only requires a guide magnetic field, along the path of motion. The final step is to trap the moving plasmoid in a separate confinement chamber. This is done by ejecting the plasmoid into a strong mirror field so that its axial motion is “trapped” by a downstream mirror.

Figure 20 illustrates the translation-trapping system. End-to-end asymmetry in the θ -pinch source (left end) is achieved by a stronger mirror on the left (LM) than the right (RM). The FIX example²¹² has an drift field section between source and confinement chamber. The confinement chamber is bounded at both ends (upstream, downstream) by quasi-steady mirror coils. The downstream mirror stops the plasmoid’ rightward translational motion and the upstream mirror stops the rebound.

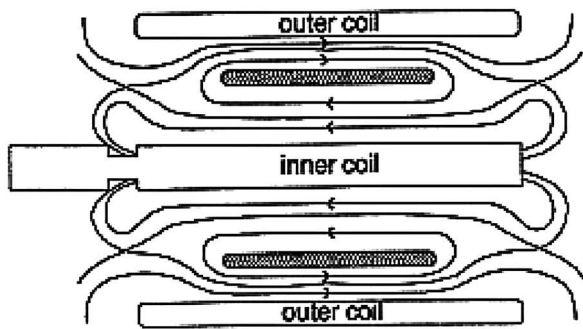


FIG. 19. Coil and plasma arrangement in the Coaxial Slow Source. Reprinted with permission from Z. A. Pietrzyk, G. C. Vlases, R. D. Brooks, K. D. Hahn, and R. Raman, Nucl. Fusion **27**, 1478 (1987). Copyright 1987, International Atomic Energy Agency.

Translation-trapping offers a major technological advantage over the traditional θ -pinch by spatially separating the source and confinement regions. The θ -pinch requires high voltage and a fast rising field, while confinement requires a quasi-steady field. Achieving both is difficult in a θ -pinch, where the field naturally falls just as rapidly as it rose, requiring a difficult “crowbar” switching operation to transition from pulsed to quasi-steady. Moreover, a separate confinement chamber more easily accommodates the needs for current drive and auxiliary heating than a θ -pinch, e.g., neutral beam ports or radio-frequency antennas. Finally, in fusion applications, insulators needed in a θ -pinch must be isolated from the nuclear environment of the confinement chamber.

It has long been known that plasmoids formed in a conical θ -pinch eject out of the end of the pinch coil.²⁷³ The conical coil structure creates an asymmetric axial magnetic force, higher on the small end than on the large end. Experiments later demonstrated the “pusher” coil technique of ejection,¹² as with a stronger “LM” than “RM” in Fig. 20. Trapping of translating plasmoids was subsequently at the Kurchatov Institute,^{12,274,275} FRX-C/T,^{276,277} OCT-L1,²⁷⁸ FIX,^{22,279} FRX-C/LSM,²⁸⁰ and TCS.^{24,89,281} An experiment is also being developed to use FRX-L as the source for MTF using translation-trapping.^{264,282} These facilities have spanned a broad range of parameters and geometries. For example, the ratio of the source coil radius R_θ to the confinement coil radius R_c has varied from as high as $R_\theta/R_c = 3/2$ on FRX-C/T (Ref. 277) to as low $R_\theta/R_c = 1/3$ on FIX.^{212,283}

b. Observations. Translation-trapping formation exhibit several striking properties. (1) *Dynamical source*, the source plasmoid is far from equilibrium since its axial ejection is more-or-less simultaneous with the formation. (2) *High-speed translation*, as high as 200 km/s in FIX,^{212,283} which is 2–3

times the speed of sound. (3) *Recovery*: despite the violence of the ejection, the plasmoid relaxes into a quiescent state, even though there may be transient contact with the downstream end of the chamber.^{24,103} (4) *Thermalization*, while early results did not observe recovery of the kinetic energy of translation,^{274,277} later experiments showed that it converted efficiently to thermal energy.^{212,278,283} The critical factor was the speed of translation: at supersonic speeds, a shock-like phenomena occurred in the plasmoid when it rebounded from the downstream mirror, converting a large fraction of the kinetic energy into ion thermal energy. (5) *Toroidal field (TF) generation* was observed in early experiments⁵ and more recently.^{22,103,261} In one case, the TF decayed rapidly,²⁶¹ but in the others, it persisted with a decay time comparable to the FRC lifetime. The toroidal flux could be as high as two to three times the poloidal flux because of the large elongation.^{22,103} (6) *Flux conversion*, the conversion of toroidal to poloidal flux has been observed in the transition to equilibrium.^{89,103} This may reflect a relaxation process, which is addressed further in Sec. IV. The MOQUI (Ref. 100) and more recently the LamyRidge¹⁷ codes have been used to simulate translation-trapping formation. Quasi-analytic formation models also continue in use.²⁸⁴

c. Technology. The ejection speed depends on such things as the effective “cone” angle in the source as well as differences between the source magnetic field and the bias field in the confinement chamber. The ejection speed can be increased by a sequenced-coil technique in which sections of the coil are energized in sequence so as to produce a strong axial magnetic force. This was applied in transition coil section between the θ -pinch source and the confinement chamber^{33,281,285} and in the θ -pinch coil itself.^{16,17} Another technique employs a “dynamical mirror” to slow the translation speed;²⁸⁶ this might be used to mitigate the violence of the collision with and downstream mirror. Other techniques exploit the high translational speed of the plasmoid as it enters the confinement chamber. The relative speed between the plasmoid and stationary background neutrals there is high enough that the neutrals collide with the plasmoid like an injected neutral beam. The associated heating and particle injection effect has been investigated in the NUCTE-III facility.^{238,287}

E. Collision merging

a. Introduction. Another formation method that uses θ -pinch sources is collision merging. Sources at each end of the confinement chamber are fired simultaneously. The ejected plasmoids collide at the center of the chamber. Upon

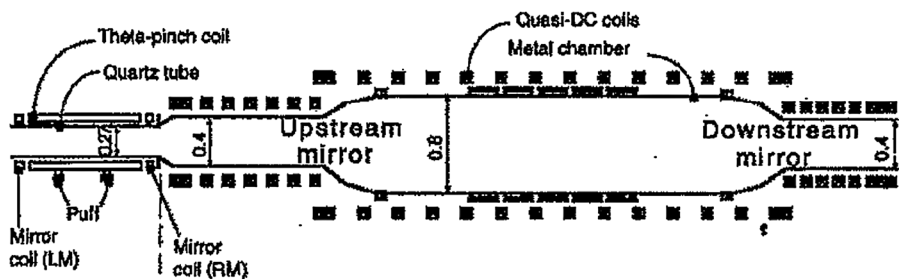


FIG. 20. Translation trapping system in the FIX experiment. Reprinted with permission from H. Himura, S. Okada, S. Sugimoto, and S. Goto, Phys. Plasmas **2**, 191 (1995). Copyright 1995, American Institute of Physics.

collision they merge and relax quickly to a quiescent FRC. Figure 21 shows this arrangement in the C-2 facility.¹⁶

The idea of merging colliding plasmoids is an old one. An early experiment studied it,²⁸⁸ although the diagnostics were insufficient to determine if a quiescent merged equilibrium resulted. Recently, collision-merging has been verified in the IPA (Ref. 35) and C-2 (Ref. 16) facilities. The C-2 experiments^{16,17} have achieved a separatrix radius $R_s \sim 0.4$ m, the largest yet in an FRC; the poloidal flux $\phi_p \sim 15$ mWb was comparable to that in the largest θ -pinch FRC despite a somewhat lower external field. The NIMROD (Ref. 289) and LamyRidge¹⁷ codes have simulated collision merging.

b. Observations. (1) *Translation speed*: high speeds ~ 80 – 250 km/s have been observed, up to four times the Alfvén speed.¹⁶ (2) *Merging*: the collision of the plasmoids led to merging into a single object in ~ 10 μ s, rapidly settling into a quiescent equilibrium in ~ 20 μ s. (3) *Rethermalization*: the kinetic energy of the translating plasmoids converted with high-efficiency (60%–70%) into thermal energy in the equilibrium, with most of the remainder going into “pumping up” the external magnetic field. This energy went almost entirely into the ions, leading to a relatively high temperature ratio $T_i/T_e \sim 4.5 - 5.5$. Unlike in translation-trapping, the rethermalization efficiency depends little on the merging speed. (4) *Poloidal flux*: the achieved ϕ_p increases notably with translation speed. (5) *Flux amplification*: based on rigid-rotor estimates and limited internal field probing the poloidal flux between ejection and equilibrium may increase by as much as factor of 10.¹⁷

F. Spheromak merging

Formation by θ -pinch methods implies high-voltages, a demand that grows more acute in larger, fusion-relevant systems. This issue fostered the search for more technology-friendly methods, the most successful of which has been “merging-spheromaks.” Two spheromaks with a common geometric axis are separately formed; driven together they attach and coalesce. If the “parent spheromaks” have opposite helicity, i.e., opposite toroidal fields, they merge to form an FRC with high- β and little or no toroidal field. This differs from collision merging in that the plasmoids meet at relatively low speed. Originally demonstrated on a small scale,^{18,19} spheromak merging has since been applied in a number of facilities.^{18,26,28,36,48,113,131,290,291} Several variations of the basic method have appeared, as described following:

- (1) *z- θ discharge*. The simultaneous discharge of an “axial” current between two electrodes and an θ current inductively generated by a poloidal field coil creates a spheromak.

If two such systems are mounted at opposite ends of the vacuum vessel, their respective spheromaks merge in the center, as illustrated in Fig. 22. The Tokyo Spheromak 3 (TS-3) experiment^{18,113,131} employs this method.

- (2) *Coaxial guns*. The parent spheromaks can also be formed by coaxial plasma “guns.” In each gun a “radial” current is drawn across the annulus between the inner and outer cylinders (electrodes) in the presence of a poloidal field formed by an θ -current coil located inside the inner cylinder. The resulting spheromak ejects naturally from the end of the inner electrode. When two such guns are mounted at opposite ends of a vacuum vessel, with codirectional “radial” and “ θ ” currents, the parent spheromaks meet at the center and merge to form an FRC, as illustrated in Fig. 23. The Swarthmore Spheromak eXperiment (SSX) (Ref. 26) applies this method.
- (3) *Flux cores*. Spheromaks can also be formed by a purely inductive approach using toroidal flux cores containing both toroidal and poloidal windings. Flux cores mounted at each end of the chamber produce spheromaks that approach and merge into an FRC if the toroidal fields in each are opposite. The arrangement resembles that in Fig. 22 except that the flux cores replace the “poloidal field coils” and “z”-current electrodes. The Tokyo Spheromak 4 (TS-4) (Ref. 113,28) and the Magnetic Reconnection eXperiment (MRX) (Ref. 48) facilities apply flux-core formation.
- (4) *Inductively-driven spheromak*. If the parent spheromaks have the same helicity (co-directional toroidal field), they merge at the center to form a spheromak. The central column (as in Fig. 22) can be equipped with solenoid (Ohmic) windings. After the spheromak is formed, ramping up the solenoid current raises the toroidal current in the spheromak as the same time as its poloidal current is decaying; this can lead to an FRC. Applied in MRX, this proved to be a robust and effective formation method.^{290,291} A proposed variation in this approach employs a single flux core and a truncated central flux core, as illustrated in Fig. 24.^{291,292} The FRC can then be translated from the source region by reducing the “holding coil” current.

a. Observations. Merging-spheromak FRCs exhibit characteristic properties. (1) *Oblate*. The method naturally forms *oblate* FRCs in marked contrast to the prolate (elongated) FRCs formed by θ -pinch-based methods. (2) *Bifurcation*. If the helicities of the “parent” spheromaks are opposite with the same magnitude, then merging produces an FRC. However, the magnitudes of the two helicities may differ by as much as 20% and still form an FRC; although a net

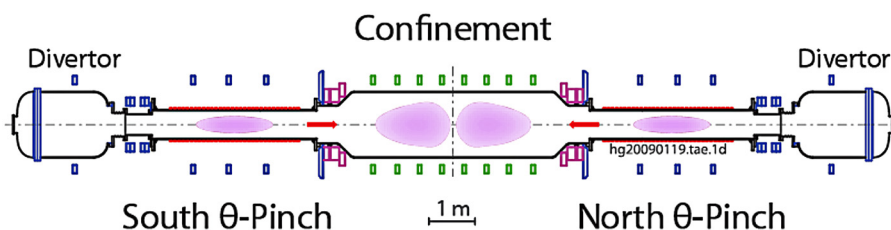


FIG. 21. (Color online) Schematic of the C-2 facility. Reprinted with permission from M. W. Binderbauer, H. Y. Guo, M. Tuszewski *et al.*, Phys. Rev. Lett. **105**, 045003 (2010). Copyright 2010, The American Physical Society.

toroidal field appears initially, it rapidly decays to form an FRC. If the two helicities differ by a larger amount, then the merged object remains a spheromak. This bifurcation was readily apparent in TS-3.^{131,203} However, in the larger higher- S^* TS-4 facility, the relative helicity margin producing an FRC was much narrower.^{28,29} (3) *Intrinsic heating.* The reconnection in the merging process strongly accelerates or “sling shots” ions away from the reconnection point. This Alfvénic motion thermalizes, converting the initial toroidal field energy into ion thermal energy.^{293,294}

b. Modelling. Two early two-dimensional (2D) MHD simulations addressed the merging-spheromak start-up method and captured many of its features.^{18,295} More recently, 2D and 3D MHD simulations have modeled start-up in SSX.¹⁰² These predict large flows approaching the Alfvén speed, significant ohmic and (ion) viscous heating. While the resistivity has little effect on the duration of reconnection, the toroidal field structure depends strongly on the viscosity, and especially so for the highly viscous SSX plasma. The 3D simulations differ little from the 2D until, of course, the appearance of tilting. Recent hybrid simulations have also addressed SSX start-up.²⁹⁶ Early results resemble the MHD simulations but thus far have only examined the early stage of the merging.

G. Rotating magnetic fields

a. Introduction. Another possibility for low-voltage FRC formation is a rotating magnetic field (RMF).⁵ This method has a long history; since the early 1960s devices call “rotamaks” were investigated in Australia (e.g., see the review in Ref. 297). Rotamak plasmas resemble FRCs in that the plasma current creates a reversed magnetic field structure. On the other hand, they differ in other, important ways that bear on the relevance of RMF to magnetic fusion: (1) *no separatrix*, i.e., no clear open-field region separating the plasma from the vacuum chamber wall; (2) *relatively cold*, $T_i + T_e < 20$ eV; (3) *low magnetic field*, $B \sim 20$ G and density ($n \sim 10^{18} \text{ m}^{-3}$); and (4) *partial ionization*, i.e., the plasma exists in a somewhat denser bath of neutrals. Even so, theo-

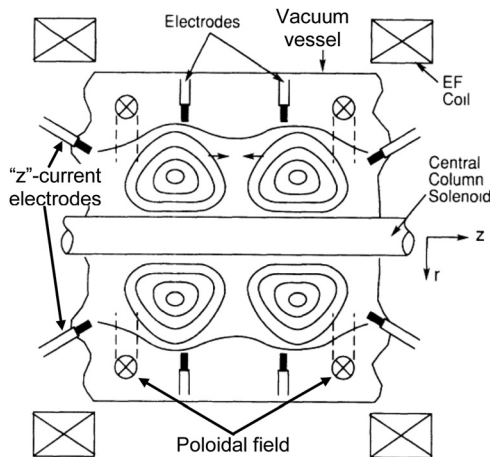


FIG. 22. Merging-spheromaks by the z - θ discharge method. Reprinted with permission from M. Yamada, Y. Ono, A. Hayakawa, and M. Katsurai, Phys. Rev. Lett. **65**, 721 (1990). Copyright 1990, The American Physical Society.

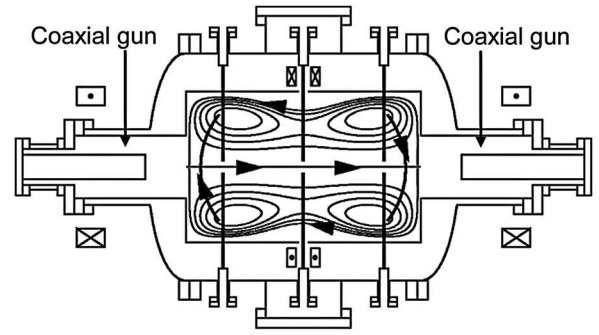


FIG. 23. Merging-spheromaks using coaxial guns. The three vertical lines mark magnetic probe locations in SSX. Reprinted with permission from C. D. Cothran, A. Falk, A. Fefferman, M. Landreman, M. R. Brown, and M. J. Schaffer, Phys. Plasmas **10**, 1748 (2003). Copyright 2003, American Institute of Physics.

retical studies in the early 1980s addressed the prospects of RMF fusion-relevant plasmas. The late 1990s saw the initiation of an experimental programs to adapt RMF to FRC-like conditions: the Star Thrust eXperiment (STX) (Refs. 298, 299) and the Translation Confinement Sustainment (TCS) experiment.^{24,25} Since then other facilities have applied RMF formation as well, FIX,³⁰⁰ PFRC,³² and TCS-U.^{30,31} Meanwhile, experiments on traditional rotamaks have continued.^{107,126,301} Figure 25 illustrates the basic antenna system for generating an RMF field. The two sets of antennas, properly sequenced, create a rotating transverse field with a dipole structure. RMF physics is discussed in more detail in Sec. VII, primarily on its ability to sustain the current. Here, attention focuses on formation per se.

b. Observations. Experiments to form FRCs using RMF have shown several advances over rotamaks. (1) High-elongation $E \sim 5$ compared to spherical ($E \sim 1$). (2) A layer of external magnetic flux, although the separatrix remains fairly close to the wall $R_s/R_c \sim 0.9$. (3) Moderate temperatures: T_i up to 150–200 eV.^{30,32} (4) High confinement magnetic field: B_e up to 25 mT (compared to < 2 mT);³⁰ importantly the ratio B_{RMF}/B_e was as low as 1/12 compared to unity in rotamaks.³² (5) Nearly complete ionization, with ionization percentages as high as 90%.²⁴¹ RMF formation has been modeled by two-dimensional MHD simulations as will be discussed in Sec. VII.

H. Energetic particle injection

Creating an FRC by injecting a component of energetic particles is an old idea, dating to the Astron concept of the late 1950s and 1960s.¹ Early investigations were reviewed in 1982.⁴ These include the field-reversed mirror (FRM), which uses neutral-beam injection (NBI),³⁰² and field-reversing electron rings.³⁰³ The FRM concept is an extension of magnetic mirrors where the injected neutral beam both heats the plasma and raises its β by driving a strong diamagnetic current.³⁰² The goal of the FRM is to inject sufficient beam current to reverse the magnetic field. Experiments in the 2XIIIB mirror facility produced a large diamagnetic current and high density ($> 10^{20} \text{ m}^{-3}$) but failed to achieve complete field reversal. It remains unclear whether this was due to inadequate neutral beam current, as suggested in Ref. 302 or

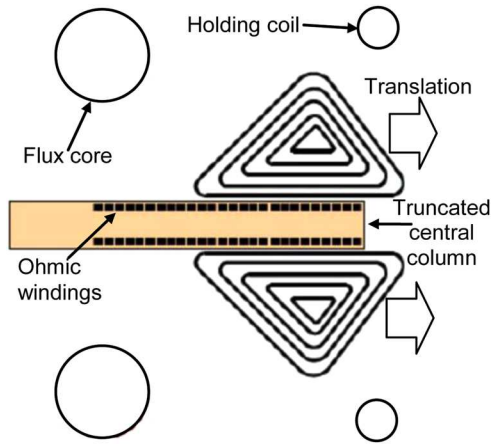


FIG. 24. (Color online) Proposed inductive-spheromak start-up with translation. Reprinted with permission from S. P. Gerhardt, E. V. Belova, M. Yamada, H. Ji, M. Inomoto, C. M. Jacobson, R. Maqueda, B. McGeehan, and Y. Ren, *Phys. Plasmas* **15**, 022503 (2008). Copyright 2008, American Institute of Physics.

because of instabilities associated with the near field-reversed state. About the same time, injection of a highly-relativistic (2 MeV) electrons into a solenoid magnet created an electron ring with sufficient current to produce full field reversal $\delta B/B_0 \sim 1.7$.³⁰³

Later work focused on ion ring injection. The Ion Ring Experiment (IREX) injected a proton beam from a ion diode through a magnetic cusp to create a proton ring,³⁰⁴ creating a measurable diamagnetic current, and $\delta B/B_0 \sim 0.04$. This raised the possibility of fully reversing the magnetic field⁶⁶ and motivated the Field-reversing Ion Ring Experiment FIREX (Ref. 305) and a simulation effort in parallel.³⁰⁶ However, achieving field reversal by this means proved elusive because of instabilities, and the effort was abandoned.³⁰⁷

VII. SUSTAINMENT

A. Introduction

Fusion energy follows two philosophically different approaches: (1) magnetic fusion, a sustained steady state with relatively low density and (2) inertial confinement fusion

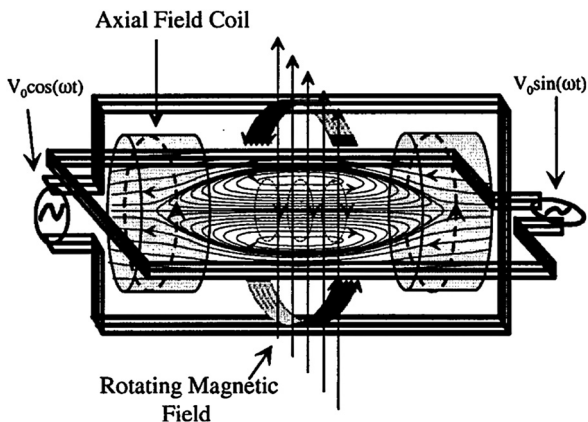


FIG. 25. RMF antenna system. Reprinted with permission from J. T. Slough and K. E. Miller, *Phys. Rev. Lett.* **85**, 1444 (2000). Copyright 2000, The American Physical Society.

(ICF), a transient state at quite high density. In addition, there is a “bridge” approach, magnetized target fusion (MTF), which employs magnetic confinement, but in a transient state and at intermediate densities. The FRC concept has one foot in each of the magnetic fusion and MTF worlds. The emphasis of this chapter is on the magnetic-fusion approach in which sustainment is an essential issue. Broadly speaking, sustainment includes several elements: (1) *current drive*, i.e., sustaining the plasma current that creates the FRC configuration; (2) *refueling*, i.e., restoring lost particles; and (3) *heating*, i.e., adding energy to maintain the temperature as a desired level for fusion burning. Prior to 1988 (Ref. 5), essentially all FRC research was focused on a transient plasma as formed in a θ -pinch with little thought on how to sustain it. The only exceptions were the field-reversed-mirror concept and preliminary thoughts on how to adapt the rotamak current drive method in the FRC context. Since then, however, a great deal of attention has focused on FRC current drive; indeed, this was the primary purpose of the TCS experiments.^{24,25,240}

The question of sustainment is as much technological as it is “plasma physics” per se. As such, it along with “formation” have been left to the last two sections of this review. The remainder of this section is organized according to the various aspects of sustainment: (B) current drive, (C) refueling, and (D) heating.

B. Current-drive

1. Rotating magnetic fields (RMF)

The essence of RMF is to generate a transverse magnetic field (B_ω) using multipole-like antennas. The antenna current is oscillated in such a way that the B_ω rotates. This RMF can drive a current if it penetrates the plasma, if the electrons are magnetized (in the B_ω), and the ions are not so that the RMF drives a net current. RMF has been reviewed in the original *rotamak* context,²⁹⁷ in terms of the theory,³⁰⁸ and recently in a succinct summary.²⁹² Traditionally, rotamaks employ a spherical chamber and produce a steady but relatively cool plasma ($T_e \sim 5$ eV) in more-or-less immediate contact with the wall. The possibility of adapting this to an FRC was considered in the 1980s.⁵ Adapting to FRCs requires several differences from a rotamak: (1) reducing the RMF field to a small fraction of the equilibrium field; (2) creating an external “vertical field” layer isolating the plasma from cold boundaries; (3) achievement of hotter, more dense conditions relevant to fusion, e.g. $T > 100$ eV, $n > 10^{20}$ cm⁻³; and (4) preferably, achieving an elongated separatrix.

a. Experiments. Several FRC-relevant experiments employing the RMF technique have been undertaken since 2000. A summary of these and some of their achievements follows: The Star Thrust eXperiment (STX) demonstrated higher toroidal current (and hence higher magnetic field) than in rotamaks and achieved $T_e \sim 40$ eV.^{298,299} The TCS experiments achieved $T_i \sim 50$ eV; $n_e \sim 10^{19}$ m⁻³, $B_e \sim 20$ mT, in the largest FRC at the time, $R_s = 35$ cm.²⁴ TCS was operated at more than one RMF frequency at a time²⁵ yielded detailed internal profile and frequency spectrum using internal magnetic probes⁹¹ and demonstrated long-pulse sustainment, with

lifetimes up to 10 ms.⁹¹ The PFRC experiments³² produced a smaller ($R_s \sim 3$ cm) but much hotter plasma ($T_e \sim 230$ eV) at a density of $\sim 2 \times 10^{18} \text{ m}^{-3}$. The TCS-U experiments³¹ introduced several plasma hygiene techniques to reduce impurities, including moderate wall baking and glow-discharge cleaning. In addition, it added flux-conserving rings to provide improved equilibrium control.³⁰⁹ TCS-U achieved $T_e > 200$ eV (Refs. 30 and 240) increased B_e/B_ω from 3 to 6 (a measure of current drive efficiency) with B_e up to 0.03 T.²⁴⁰ Later, experiments on TCS-U introduced recycling controls (silicization, titanium gettering).²⁴¹ The FIX facility was the host for a series of RMF experiments in which the internal structure was explored in detail using electrostatic probes.³¹⁰ The Kansai RMF experiment explored penetration by the RMF as well as its influence on radial force balance.³⁰¹

Besides the parameters achieved, several important properties emerged from the experiments. Steady state was achieved even though the RMF field was a skin effect, i.e., it only penetrated a short distance ($1/5 - 1/10 \times R_s$) inside the separatrix.^{298,299} This was attributed to an intermittent reconnection process transferring the edge current inwards.³¹¹ Long-pulse operation exhibited the spontaneous appearance of a new high-performance mode with even shallower penetration, larger poloidal flux generation, and lower resistivity.⁹¹ RMF plasmas exhibited much more peaked density profiles than other FRCs, with separatrix-to-maximum density ratios of ~ 0.2 compared with a typical $0.5 - 0.7$ in θ -pinch-based FRCs.²⁴⁰ The typical ion rotation rate was limited to $\sim 1/5 \omega_{RMF}$ (RMF frequency).²⁵ Current drive using a quadrupole was demonstrated, yielding similar performance to the usual dipole RMF system.³¹² “Odd parity” antennas, split at the midplane were also applied.³²

a. Theory. Modelling of RMF physics has advanced in recent years, to include factors missing from earlier theories. These include radial flow and radial current effects,^{313,314} RMF penetration,^{315,316} the concept of balance of *torques* (RMF vs resistive),³¹⁶ ion physics and transient ion spin-up,³¹⁷ radial force balance correction from RMF fields,³¹⁷ rotation rate stability,³¹⁸ as well as the development of practical scaling laws.³⁰⁸ Various innovations have also been proposed, including “odd parity” antennas³¹⁹ and a two-frequency-driving system with one frequency tuned to the electrons and one to the ions.³²⁰ Simulations of RMF have also been developed. A full MHD simulation included ion motion in two dimensions (r, θ), some three-dimensional effects³²¹ and neutral ionization.³²² A recent fully-three-dimensional (3D) MHD simulation examined the axial structure and the effect of end shorting.³²³ A 3D “particle-in-cell” code examined particle orbits and energetic effects.³²⁴

b. Liabilities of RMF current drive. RMF is fundamentally an electron current drive method. However, if the ions spin up, they can cancel the electron current and render the current drive ineffective. This was originally recognized in the context of current drive by neutral beam injection.¹⁵ The theory of transient ion spinup showed that it cancels the current drive effect in a relatively short time (e.g., 100 μs) unless the ion rotation is retarded in some way, e.g., by drag

on neutrals.³¹⁷ Simulations suggested the same thing: if the ions are allowed to rotate, an ionization source (i.e., the presence of neutrals) was required in order for current drive to be sustained.³²² Experiments confirm this. Elimination of recycling by gettering, thereby removing the surrounding “bath” of neutrals, caused the current drive to turn off after $\sim 100 \mu\text{s}$ unless additional neutrals were continuously puffed into the chamber.²⁴¹ Investigation of rotation rates and various ion drag effects showed that drag against background neutrals is the primary reason why RMF current drive can work.³²⁵ This calls into question the relevance of RMF current drive in a fusion environment where the neutral density would be very small. Other liabilities have emerged as well, although less critically: the separatrix naturally rests quite close to the wall ($R_s/R_w \sim 0.9$), the efficiency factor B_e/B_ω has remained quite limited (typically five or less) and the density has been limited to about 10^{19} m^{-3} or so.

2. Neutral beam injection (NBI)

NBI was always a central element in the proposed field-reversed-mirror concept, both for current drive and heating. It has re-emerged recently as an equally important element of FRC fusion concepts.^{326–330} While several theoretical treatments have addressed NBI in FRCs, only one experiment has been reported.

a. Experiment. A modest neutral beam (8–14 kV, 17 A) was injected in the FIX experiment.²⁵⁵ Because of the modest poloidal flux in FIX, trapping of the “beam ions” produced by ionization is unlikely for tangential injection. This difficulty was managed by highly oblique injection, i.e., aiming the beam from one end at a small angle (19°) to the geometric axis. The large applied mirror ratio in FIX (up to 14) also contributed to good beam-ion confinement. This and later experiments^{166,300,331} showed that NBI heated the electrons (only) and gave them an oddly inverted temperature profile with a local minimum near the O-point. When the mirror ratio was high enough to effectively trap the beam ions, the confinement time was increased by as much as 50%.³⁰⁰ About 50% of the nb energy was absorbed in the plasma.³³¹

b. Theory. Confinement of “beam ions” produced by tangential NBI is difficult unless the poloidal flux ϕ_p of the plasma is high enough. For tangential beam injection at the magnetic axis, the requirement that the beam orbit not extend outside the separatrix is³³⁰

$$\phi_p \geq 2\pi(R_s/q_b)\sqrt{2m_bW_b}, \quad (28)$$

where m_b , q_b , and W_b are the beam particle mass, charge, and energy. This is a fairly stringent requirement: e.g., injection of a 15 kV hydrogen beam into a moderately larger FRC ($R_s = 0.25$ m) requires $\phi_p \geq 27$ mWb, which is about double the values achieved on the LSX facility, the largest-flux FRC facility to date. It is likely that a significant fraction of beam ions will have orbits extending outside the separatrix.

Monte Carlo simulations have considered tangential NBI at the midplane, showing that high beam current tends

to “dig a hole” in the equilibrium, causing a concentration of both plasma and beam ions near the midplane.³²⁸ Later, TCS-relevant simulations in the presence of an RMF considered both tangential and oblique injection.³²⁹ While the RMF degraded beam confinement, good current-drive efficiency was possible for shallow RMF penetration. The suggestion was offered that the torque applied by the nb might balance the RMF torque. Other Monte-Carlo simulations considered highly oblique injection relevant to the FIX experiment. These highlighted beam-ion confinement issues, suggesting that all fast ions can be lost either by wall collisions or end loss.³³² This is attributed to magnetic-moment breaking near the O-point and in the steep density region at the edge.³³³ As a result, a fairly low beam power absorption of $\sim 10\%$ was predicted for FIX. This is not surprising given the estimated poloidal flux $\phi_p < 1$ mWb.

c. Liabilities of NBI current drive. The issue of (1) *beam-ion confinement* has been raised, but other possible difficulties must also be addressed before efficient NBI can be assured. (2) The question of *current cancellation* was raised early,¹⁵ although it was suggested that weakly breaking the toroidal symmetry would allow electrical contact with external structure and provide the needed torque on the electrons. This may occur naturally from surface-breaking turbulence.⁸³ (3) On the other hand, *symmetry-breaking fields* may exacerbate beam losses. This appeared not to be a severe problem for modest RMF fields,³²⁹ but the more chaotic turbulent field fluctuations might serve to expel beam ions. On the other hand, the effect of such fluctuations may be averaged out over the orbits of high-energy ions and have little effect on their confinement.⁶⁸ (4) A potentially severe liability is the charge-exchange loss of beam ions due to collisions with neutrals outside the separatrix. This places a fairly stringent restriction on the neutral density outside the separatrix; 90% beam coupling to the FRC requires $n_n < 10^{17} \text{ m}^{-3}$.^{330,334} Keeping the surrounding population of neutrals very low may conflict with other factors, e.g., any kind of gas-puff refueling, and possibly RMF, which may require neutrals to be effective.

3. Other current drive methods

Although inductive current drive methods cannot be sustained indefinitely, they can be used to ramp up the current, at least initially. *Axial compression* by raising the field at the ends served to squeeze the FRC axially,³³⁵ shorten the length and increase the separatrix radius.^{104,105} A central column containing a *central solenoid* (Re Figs. 19,22,24) has inductively driven the current on TS-3,³³⁶ TS-4,³³⁷ and MRX.^{292,338} Other methods for driving the current have also been proposed, including repetitive (intermittent) spheromak merging,¹¹³ the current-maintaining effect of the thermal force,³³⁹ and the current maintaining effect of energetic fusion products.^{340,341}

C. Heating and refueling

a. Experiment. Existing formation techniques achieve temperatures up to 1–2 keV, but higher temperature is neces-

sary for fusion energy. Various methods have been explored for both initial and ongoing heating. The simplest is compression heating by raising the external magnetic field. On FRX-C/LSM, this increased T_i from 0.4 to 1.5 keV,²⁸⁰ although it produced an accompanying degradation in confinement. Other methods have also been proposed. A series of experiments explored low-frequency wave heating in the FIX facility.^{166,342–344} An oscillating magnetic pulse was excited by an antenna consisting of two single-turn coils at a frequency somewhat below the ion cyclotron frequency at the separatrix. This produced an increase of the plasma energy ($\sim 20\%$), mostly in the ions. Internal magnetic probes measured a torsional Alfvén wave with an amplitude $\delta B_\theta \sim B_e/20$. A later study addressed the wave propagation characteristics observing two distinct wave modes.³⁴⁵ A low $k_{||} \sim 2.5 \text{ m}^{-1}$ wave propagated in a broad region extending outside the separatrix and exhibited dissipation reminiscent of resistive damping. A high $k_{||} \sim 7.0 \text{ m}^{-1}$ wave inside the separatrix was strongly damped, probably by transit-time magnetic pumping.

b. Theory. Heating by energetic NBI is well known in magnetic confinement systems and has been explored theoretically in the FRC context.^{4,70,333,346} Translation of an FRC at high speed into a neutral background produces a similar effect. The speed of the neutrals relative to the FRC is large enough that they act as a diffuse neutral beam that provides some initial heating.³⁴⁷ Several studies have addressed the heating potential of RMF. In most such experiments, the RMF power density is fairly modest, the exception being the relatively small PFRC.³² However, RMF near the ion cyclotron frequency can generate resonance heating.³⁴⁸ Further, stochastic orbit effects have the potential of strong ion heating in the presence of odd-parity RMF antenna structure.^{78,349–351}

c. Refueling. Gas refueling can be produced in the presence of RMF, which has a strong ionization and inward “pumping” effect on the plasma. Indeed, experiments on TCS-U showed that a continuous supply of neutral was necessary for the current drive effect to persist.²⁴¹ Transient refueling, i.e., a one-time increase in the particle inventory was accomplished on NUCTE-III by translation into a neutral background.^{287,347}

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APPENDIX: ESSENTIAL FRC TERMINOLOGY

FRC research employs a somewhat specialized terminology in some respects, as summarized here.

(r, θ, z) = cylindrical coordinate system usually used; $z = 0$ = midplane, usually a plane of symmetry.

B_e = external magnetic field; used in elongated FRCs where it is uniform outside the separatrix.

n_m = maximum ion density, near the O-point; $Z \approx 1$ so that the maximum electron density is similar.

T_t = total temperature (ion + electron); the maximum value is usually quoted.

R_s = separatrix radius at the midplane (Re Fig. 2).

R_0 = magnetic axis radius; roughly $R_s/\sqrt{2}$ in elongated FRCs (Re Fig. 2).

R_c = radius of conducting wall (Re Fig. 2).

L_s = separatrix length (Re Fig. 2).

$\rho_{i0} = (m_i k T_i)^{1/2} / e B_e$ = reference ion larmor radius; based on external magnetic field.

(c/ω_{pi}) = ion inertial length (ion skin depth); $\omega_{pi} = (m_i / \mu_0 e^2 n_m)^{1/2}$ is the ion plasma frequency

$V_{A0} = B_e / (\mu_0 m_i n_m)^{1/2}$ = reference Alfvén speed based on external field and maximum density

$\Delta\phi$ = excluded flux (magnetic flux excluded by the plasma)

$R_{\Delta\phi}$ = excluded flux radius, $(\Delta\phi / B_e \pi)^{1/2}$; approximately equal to the separatrix radius.

ϕ_p = internal poloidal flux.

$S_* = R_s / (c/\omega_{pi})$ = radius parameter; its inverse measures kinetic and two-fluid effects.

$\bar{s} = \int_{R_0}^{R_s} r dr / R_s \rho_i$ = alternate radius parameter; based on the local ion Larmor radius ρ_i .

$S = R_0 / \rho_{i0}$ = alternate radius parameter.

$E = L_s / 2R_s$ = separatrix elongation: oblate $E < 1$; prolate or elongated $E > 1$.

$X_s = R_s / R_c$ = separatrix radius / coil radius; affects the amount of poloidal flux.

τ_N = particle confinement time, generally the decay time of the particle inventory.

τ_ϕ = poloidal flux decay time.

τ_{Ep} : decay time of plasma energy. Typically it is $1/2 \times \tau_N$.

$\beta = nkT_i / (B_e^2 / 2\mu_0)$ = beta value based on the external magnetic field; an “engineering β .”

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¹⁰The “ S ” parameter is defined as $S \equiv R_0 / \rho_{i0}$, where $\rho_{i0} \equiv (m_i k T_i)^{1/2} / e B_e$ is the ion Larmor radius based on the external magnetic field. For typical elongated FRCs with $T_i \sim 2T_e$, $S \sim 1.3S_*$. The “ $\bar{s}$ ” parameter is defined as $\bar{s} \equiv \int r dr / \rho_i R_s$, where the integral is over the minor radius (R_0 to R_s). It differs from S in that it uses an integrated average of the local gyroradius. For typical profiles, $\bar{s} \approx X_s S / 5$. The advantage of the S_* over these alternate definitions is that it can be inferred directly from measurements without any guesswork about the ion temperature.

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